Radiative Transfer

A graduated course for pre-master students, with LTE solutions, worked examples and exam questions & answers

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Formula Sheet — Radiative Transfer (for revision and the exam)
Radiation field (Lecture 2)
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Prepared for: students entering an Astrophysics master's programme

Based on: A670 *Advanced Astrophysics* Lecture 1 · W. Hu, *Astro 305 Notes, Set 1* · MIT 8.901 (2019) Lecture 7 · G. W. Collins II, *Fundamentals of Stellar Astrophysics*, Ch. 9 · S. Chandrasekhar, *Radiative Transfer* (1950)

Code: all figures and numbers are reproducible with the Python scripts in `scripts/` (`lte_solvers.py`, `make_figures.py`, `worked_examples.py`).

Lecture 1 — Introduction: How Is the Radiative Transfer Equation Used in Astrophysics?

> **Level:** pre-master → Astrophysics master entry. **Sources:** A670 *Advanced Astrophysics* Lecture 1 (course outline); W. Hu, *Astro 305 Lecture Notes, Set 1*; MIT 8.901 (2019) Lecture 7; G. W. Collins II, *The Fundamentals of Stellar Astrophysics*, Ch. 9; S. Chandrasekhar, *Radiative Transfer* (1950), Ch. I.

Learning goals

By the end of this lecture you should be able to:

1. State in one sentence what radiative transfer (RT) is and why almost all of astrophysics depends on it.
2. Write the one-dimensional radiative transfer equation (RTE) and name every symbol in it.
3. Distinguish **thermodynamic equilibrium (TE)**, **local thermodynamic equilibrium (LTE)** and **non-LTE (NLTE)** in words.
4. List the main astrophysical applications of the RTE.

1.1 What is radiative transfer?

Almost everything we know about the Universe beyond the Solar System comes to us as **photons**. Before they reach our telescopes these photons are **emitted**, **absorbed** and **scattered** by atoms, ions, molecules, free electrons and dust grains. *Radiative transfer* is the theory of how a beam of radiation changes as it passes through matter. It is the link between:

- the **physical state** of the gas (temperature T, density ρ, chemical composition, velocity field), and
- the **observable** quantity: the spectrum I_ν that we record.

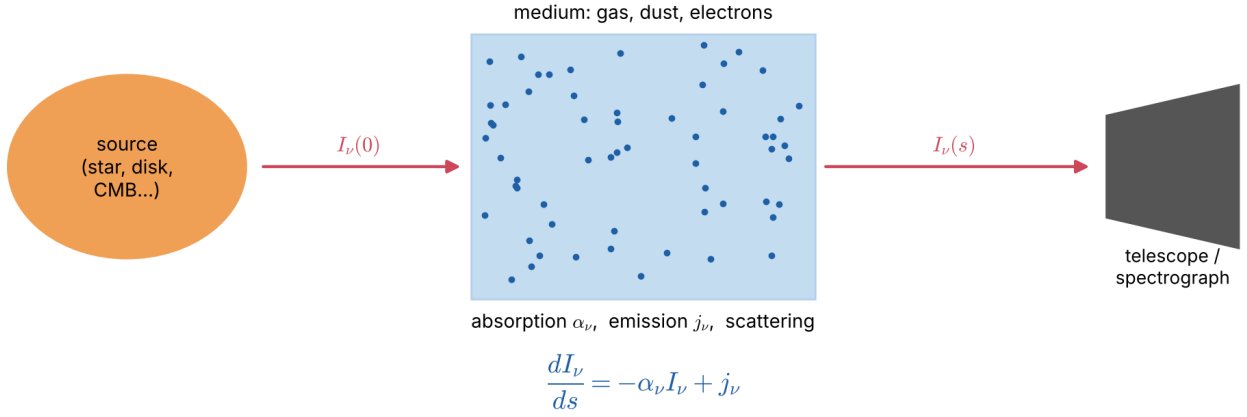


Figure 1.1 — The radiative transfer problem. A beam with intensity $I_\nu(0)$ enters a medium. Absorption (α_ν) removes energy from the beam, emission (j_ν) adds energy, and scattering does both by redirecting photons. We observe $I_\nu(s)$.

Two complementary problems:

Problem	Given	Find	Example
Forward (modelling)	$T(z)$, $\rho(z)$, composition	emergent spectrum I_ν	model stellar atmosphere, synthetic spectrum
Inverse (diagnostics)	observed I_ν	T, ρ, abundances	abundance analysis, limb-darkening inversion

1.2 The fundamental equation (preview)

Along a ray, with path length s:

$$\boxed{\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu}$$

Symbol	Name	SI units
I_ν	specific intensity	W m^{-2} Hz^{-1} sr^{-1}
$\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$	absorption (extinction) coefficient	m^{-1}
κ_ν	opacity (mass absorption coefficient)	m^2 kg^{-1}
j_ν	emission coefficient (emissivity)	W m^{-3} Hz^{-1} sr^{-1}

Physical reading. The first term is a *loss* proportional to how much light there already is, so absorption removes a fixed *fraction* per unit length. The second term is a *gain* that does not depend on I_ν; matter emits whether or not light is present.

We define the dimensionless **optical depth** $d\tau_\nu = \alpha_\nu\, ds$ and the **source function** $S_\nu = j_\nu/\alpha_\nu$:

$$\frac{dI_\nu}{d\tau_\nu} = -I_\nu + S_\nu.$$

This compact form is the one we will use throughout the course. Lectures 2–5 derive it carefully.

1.3 A short history

Year	Who	Contribution
1859	Kirchhoff	j_ν/α_ν is a universal function of T for matter in equilibrium
1900	Planck	the universal function: $B_\nu(T)$
1906	K. Schwarzschild	RTE for the solar atmosphere; radiative equilibrium explains solar limb darkening
1916–1926	Eddington	radiative equilibrium of stars; the **Eddington approximation** $K = J/3$
1917	Einstein	A and B coefficients; stimulated emission
1930–34	Hopf, Milne	exact solution of the gray atmosphere (Hopf function $q(\tau)$)
1950	Chandrasekhar	*Radiative Transfer*: discrete ordinates, H-functions, polarisation
1960s–1978	Mihalas, Auer & Mihalas	NLTE model atmospheres, numerical methods
today	many groups	3D, time-dependent, Monte-Carlo, NLTE, polarised RT on supercomputers

1.4 LTE versus non-LTE: the central idea of this course

Correction to a common statement. The A670 Lecture 1 notes state that "in LTE the radiative transfer equation simplifies to $I_\nu = B_\nu(T)$". That is the condition for **full thermodynamic equilibrium (TE)**, which holds deep inside stars. In **LTE** the correct statement is

$$S_\nu = B_\nu(T_{\rm local}) \qquad \text{but in general} \qquad I_\nu \neq B_\nu .$$

The *matter* is in equilibrium at the local temperature, so it emits like a black body per unit absorption. The *radiation field* is **not** Planckian, because photons near the surface come from regions of different temperature and can escape. If I_ν were equal to B_ν everywhere, there would be no net flux and stars would not shine. Lectures 6 and 9 return to this point in detail.

	TE (stellar interior)	LTE (most photospheres)	NLTE (upper atmospheres, nebulae, ISM)
Particle velocities	Maxwell at T	Maxwell at local T	Maxwell (usually)
Excitation / ionisation	Boltzmann + Saha at T	Boltzmann + Saha at local T	from **statistical equilibrium** (rate equations)
Source function	$S_\nu = B_\nu$	$S_\nu = B_\nu(T)$	S_ν depends on J_ν (non-local)
Radiation field	$I_\nu = B_\nu$ (isotropic)	$I_\nu \neq B_\nu$ (anisotropic, carries flux)	$I_\nu \neq B_\nu$
What sets populations?	—	collisions dominate	radiative rates compete with collisions

1.5 Where is the RTE used?

1. **Stellar atmospheres:** emergent spectra, $T_{\rm eff}$, $\log g$, abundances, limb darkening. Classic LTE territory (Lectures 7–10).
2. **The interstellar medium (ISM):** extinction and reddening by dust ($A_V = 1.086\,\tau_V$), 21-cm emission and absorption, molecular lines (CO), heating and cooling of clouds.
3. **H II regions and planetary nebulae:** recombination lines and free-free radio emission. These are optically thin to thick slabs (Lecture 5) and strongly NLTE.
4. **Planetary atmospheres:** greenhouse effect, remote sensing, exoplanet transmission spectra.
5. **Accretion disks, AGN and quasars:** disk spectra, Compton scattering, broad-line regions.
6. **Cosmology:** the CMB is the best black body known ($T = 2.725$ K); Thomson scattering at recombination and reionisation; Lyman-α forest absorption.
7. **Stellar interiors:** the diffusion limit of the RTE gives the radiative energy flux and the **Rosseland mean opacity** (Lecture 8).

1.6 How the course is built

#	Lecture	Key result
1	Introduction (this lecture)	RTE, TE/LTE/NLTE
2	Describing the radiation field	I_ν, J_ν, F_ν, K_ν; I_ν constant in vacuum
3	Black-body radiation	Planck, Wien, Stefan–Boltzmann, brightness temperature
4	Emission, absorption, scattering	α_ν, j_ν, cross-sections, Einstein coefficients, random walk
5	The RTE and its formal solution	$I = I_0 e^{-\tau} + S(1 - e^{-\tau})$; MIT toy models: thin H cloud, two-layer star, thermal inversion
6	Thermodynamic equilibrium and **LTE**	Kirchhoff, Saha–Boltzmann, $S_\nu = B_\nu$
7	Plane-parallel atmospheres	Schwarzschild–Milne, E_n, Eddington–Barbier
8	Moments, diffusion and scattering	Eddington approximation, Rosseland mean, $S(0) = \sqrt{\varepsilon} B$
9	**The gray LTE atmosphere: full solution**	$T^4 = \frac{3}{4} T_{\rm eff}^4 (\tau + q)$, limb darkening
10	**LTE spectral lines** and the limits of LTE	Milne–Eddington, curve of growth, NLTE
—	Final exam with full solutions	

Worked example 1.1 — Reading the RTE

A beam passes through a uniform slab of gas of thickness $L = 10^{12}$ m. The gas has $\alpha_\nu = 2 \times 10^{-12}$ m^{-1} and does not emit. What fraction of the beam emerges?

Solution. With $j_\nu = 0$, $dI_\nu/ds = -\alpha_\nu I_\nu$, so $I_\nu(L) = I_\nu(0)\, e^{-\alpha_\nu L}$. The optical depth is $\tau_\nu = \alpha_\nu L = 2$, and the transmitted fraction is $e^{-2} = 0.135$ (13.5 %). In magnitudes the extinction is $A = 2.5 \log_{10} e^{\tau} = 1.086\,\tau = 2.17$ mag.

Worked example 1.2 — Why does the Sun have a sharp edge?

The solar photosphere has a density scale height of about $H \approx 150$ km, and the opacity rises steeply with depth. Why does the solar limb look sharp even though the Sun is gaseous?

Solution. We see down to optical depth $\tau \approx 1$. Because $\alpha \propto \rho$ (and the opacity also rises with T), the optical depth changes from $\tau \approx 0.1$ to $\tau \approx 10$ within a few scale heights, i.e. a few hundred km. Seen from Earth, $1'' \approx 725$ km on the Sun, so the transition from transparent to opaque happens within less than $1''$, and the edge looks sharp. RT defines what "the surface" of a star means: the layer where $\tau \approx 2/3$ (Lecture 9).

Exam questions and answers — Lecture 1

Q1. Write the RTE along a ray. Explain the physical meaning of each term and why the absorption term is proportional to I_ν but the emission term is not.

> **A1.** $dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$. Absorption removes a fixed *fraction* $\alpha_\nu\, ds$ of the photons already present in the beam, so the loss is $\propto I_\nu$. Spontaneous emission is produced by the matter (excited atoms decay at rate A_{21} whether or not there is light), so it does not depend on I_ν. (Stimulated emission *is* $\propto I_\nu$ and is therefore counted as negative absorption inside α_ν; see Lecture 4.)

Q2. What is the difference between TE and LTE? "In LTE, $I_\nu = B_\nu$." True or false?

> **A2.** In TE, both matter and radiation are in equilibrium at one temperature: $I_\nu = J_\nu = B_\nu(T)$ is isotropic and there is no net flux. In LTE the *matter* is in equilibrium at the local temperature $T(\mathbf{r})$ (Maxwell, Boltzmann and Saha distributions hold), which gives $j_\nu = \alpha_\nu B_\nu(T)$, i.e. $S_\nu = B_\nu(T)$. The *radiation field* may differ from B_ν. The statement is **false**: in LTE $S_\nu = B_\nu$, not $I_\nu = B_\nu$. At the surface of a star, for example, $I_\nu(\mu < 0) = 0$ (no incoming radiation) while $B_\nu \neq 0$.

Q3. Define optical depth. A medium has $\tau_\nu = 0.01$ at one frequency and $\tau_\nu = 100$ at another. Describe what an observer sees at each frequency.

> **A3.** $\tau_\nu = \int \alpha_\nu\, ds$ is the path length measured in photon mean free paths. At $\tau_\nu = 0.01$ (optically thin), 99 % of the background light passes through; the medium adds $\approx \tau_\nu S_\nu$, so we see through it, and its own emission is weak and proportional to its column density. At $\tau_\nu = 100$ (optically thick), the background is blocked (e^{-100}), and we see only the medium's own surface layers, $I_\nu \to S_\nu$ evaluated near $\tau_\nu \approx 1$.

Lecture 2 — Describing the Radiation Field: Intensity, Flux and Moments

> **Sources:** Hu, *Astro 305 Set 1* (observables, auxiliary quantities); Collins Ch. 9.2–9.3 (specific intensity, moments); Chandrasekhar Ch. I §1–6; MIT 8.901 Lec. 7.

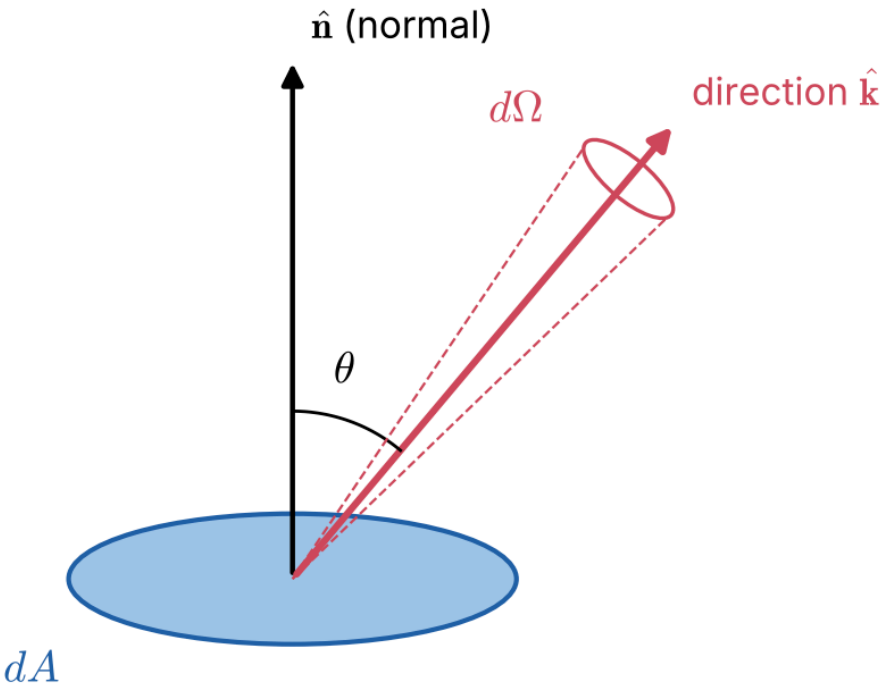
Learning goals

1. Define the specific intensity I_ν and explain why it is the fundamental quantity.
2. Derive the mean intensity J_ν, flux F_ν, energy density u_ν and radiation pressure P_ν as angular moments of I_ν.
3. Prove that I_ν is constant along a ray in empty space.
4. Use these definitions to compute fluxes from stars.

2.1 The four observables

A detector can measure four things about radiation (Hu): **energy flux**, **direction**, **frequency** and **polarisation**. Measuring the first three together defines the specific intensity. Polarisation adds the Stokes parameters Q, U, V; we use only I in this course.

2.2 Specific intensity



$$dE = I_\nu \cos\theta \, dA \, d\Omega \, d\nu \, dt$$

Figure 2.1 — Energy dE crosses an area dA whose normal $\hat{\mathbf{n}}$ makes angle θ with the direction of propagation $\hat{\mathbf{k}}$, into solid angle $d\Omega$, in frequency interval $d\nu$ and time dt.

$$\boxed{dE = I_\nu(\mathbf{r}, \hat{\mathbf{k}}, t)\ \cos\theta \ dA \ d\Omega \ d\nu \ dt}$$

Units: W m^{-2} Hz^{-1} sr^{-1} (cgs: erg s^{-1} cm^{-2} Hz^{-1} sr^{-1}). The factor $\cos\theta$ is the *projected* area seen by the beam.

Per unit wavelength: $I_\nu \, |d\nu| = I_\lambda \, |d\lambda|$. Since $\nu = c/\lambda$, $|d\nu/d\lambda| = c/\lambda^2$, so

$$I_\lambda = \frac{c}{\lambda^2} I_\nu, \qquad \nu I_\nu = \lambda I_\lambda .$$

Relation to the photon distribution function (Collins eq. 9.2.8). If $f(\mathbf{r}, \mathbf{p})$ is the photon phase-space density, then

$$I_\nu = \frac{h^4 \nu^3}{c^2} \, f .$$

Since f is conserved along a photon trajectory (Liouville's theorem), I_ν / ν^3 is invariant. This is why surface brightness dims as $(1+z)^{-4}$ in cosmology (Hu).

2.3 Constancy of I_ν along a ray in vacuum

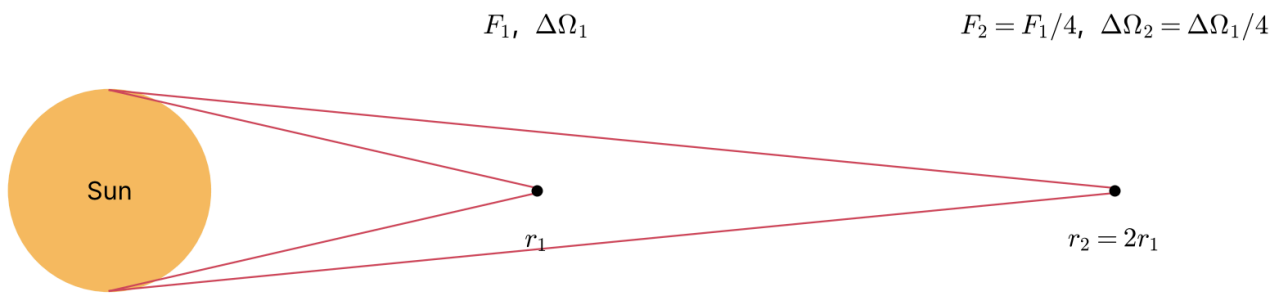


Figure 2.2 — The flux from the Sun falls as r^{-2}, but so does the solid angle it subtends. Their ratio, the intensity, is independent of distance.

Proof. Consider two small areas dA_1 and dA_2 separated by distance r along a ray, both perpendicular to it. Photons that pass through both areas cross dA_1 into solid angle $d\Omega_1 = dA_2/r^2$, and cross dA_2 from solid angle $d\Omega_2 = dA_1/r^2$. With no matter in between, energy is conserved:

$$I_{\nu,1}\, dA_1\, d\Omega_1\, d\nu\, dt = I_{\nu,2}\, dA_2\, d\Omega_2\, d\nu\, dt \;\Rightarrow\; I_{\nu,1}\frac{dA_1 dA_2}{r^2} = I_{\nu,2}\frac{dA_2 dA_1}{r^2} \;\Rightarrow\; I_{\nu,1} = I_{\nu,2}.$$

So $dI_\nu/ds = 0$ in vacuum. **Consequence:** the surface brightness of a resolved object does not depend on its distance (in a static, non-expanding space).

2.4 Angular moments of the intensity

Use spherical coordinates around a chosen axis: $d\Omega = \sin\theta\, d\theta\, d\phi = -d\mu\, d\phi$ with $\mu = \cos\theta$.

Moment	Definition	Plane-parallel form (I independent of ϕ)	Physical meaning
zeroth J_ν	$\frac{1}{4\pi}\oint I_\nu\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu\, d\mu$	mean intensity
first F_ν	$\oint I_\nu \cos\theta\, d\Omega$	$2\pi\int_{-1}^{1} I_\nu\, \mu\, d\mu$	net (physical) flux
first H_ν	$\frac{1}{4\pi}\oint I_\nu \cos\theta\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu\, \mu\, d\mu$	Eddington flux, $H_\nu = F_\nu/4\pi$
second K_ν	$\frac{1}{4\pi}\oint I_\nu \cos^2\theta\, d\Omega$	$\frac{1}{2}\int_{-1}^{1} I_\nu\, \mu^2\, d\mu$	radiation pressure, $P_\nu = 4\pi K_\nu/c$

> **Warning about flux definitions** (Collins §9.3b). Books use three different "fluxes": the physical flux F_ν (W m^{-2} Hz^{-1}), the Eddington flux $H_\nu = F_\nu/4\pi$, and the "astrophysical flux" $\mathcal{F}_\nu = F_\nu/\pi$. Always check which one a formula uses. **In this course, F is the physical flux** and $F = \sigma T_{\rm eff}^4$.

Energy density

Radiation crossing dA in time dt fills a cylinder of volume $dV = c\,dt\,dA$ (Hu). The energy in that volume travelling into $d\Omega$ is $dE = I_\nu\, dA\, d\Omega\, d\nu\, dt$, so the specific energy density per solid angle is I_ν/c. Integrating over angles:

$$u_\nu = \frac{1}{c}\oint I_\nu\, d\Omega = \frac{4\pi}{c} J_\nu, \qquad u = \int_0^\infty u_\nu\, d\nu .$$

Radiation pressure

A photon carries momentum E/c along $\hat{\mathbf{k}}$. The component normal to a surface is $(E/c)\cos\theta$, and the rate at which it crosses the surface carries another factor $\cos\theta$. For a surface that absorbs everything that hits it:

$$P_\nu = \frac{1}{c}\oint I_\nu \cos^2\theta\, d\Omega = \frac{4\pi}{c} K_\nu .$$

(A perfectly reflecting wall feels twice this from one side; Hu.)

The isotropic case

If $I_\nu = I$ is independent of direction:

$$J_\nu = I, \qquad F_\nu = 2\pi I \int_{-1}^{1} \mu\, d\mu = 0, \qquad K_\nu = \tfrac{1}{2} I \int_{-1}^{1} \mu^2 d\mu = \frac{I}{3} = \frac{J_\nu}{3}.$$

Therefore

$$\boxed{P = \tfrac{1}{3} u} \qquad \text{(equation of state of isotropic radiation)},$$

and $K_\nu = J_\nu/3$ is the basis of the **Eddington approximation** (Lecture 8).

One-sided flux

The flux leaving a surface that radiates with the same intensity I in every outward direction ($0 \le \mu \le 1$), with nothing coming in:

$$F^{+} = 2\pi \int_0^1 I\, \mu\, d\mu = \pi I .$$

This is the origin of the ubiquitous factor π: a black body has $F = \pi B = \sigma T^4$.

2.5 Flux from a uniform sphere

A star of radius R with uniform surface intensity I (no limb darkening), seen from distance r: it subtends a cone of half-angle θ_c with $\sin\theta_c = R/r$.

$$F(r) = \int_0^{2\pi}\!\!\int_0^{\theta_c} I \cos\theta \sin\theta\, d\theta\, d\phi = \pi I \sin^2\theta_c = \pi I \left(\frac{R}{r}\right)^2 .$$

At the surface ($r = R$), $F = \pi I$; far away, $F \propto r^{-2}$, while I itself is unchanged.

Worked example 2.1 — The Sun: from luminosity to intensity

Given $L_\odot = 3.828 \times 10^{26}$ W, $R_\odot = 6.957 \times 10^{8}$ m and 1 AU $= 1.496 \times 10^{11}$ m, find (a) the surface flux and $T_{\rm eff}$, (b) the mean intensity of the solar disk, (c) the solar constant, computed two ways.

(a) $F_\odot = L_\odot / (4\pi R_\odot^2) = 6.294 \times 10^{7}$ W m^{-2}. With $F = \sigma T_{\rm eff}^4$: $T_{\rm eff} = (F/\sigma)^{1/4} = 5772$ K.

(b) $\bar I = F/\pi = 2.003 \times 10^{7}$ W m^{-2} sr^{-1}.

(c) Solid angle of the Sun at 1 AU: $\Omega_\odot = \pi (R_\odot/\mathrm{AU})^2 = 6.794 \times 10^{-5}$ sr. Since I is conserved and the disk is small (so $\cos\theta \approx 1$), $F_\oplus = \bar I\, \Omega_\odot = 1361$ W m^{-2}. By the inverse-square law, $F_\odot (R_\odot/\mathrm{AU})^2 = 1361$ W m^{-2}, the same.

Worked example 2.2 — Radiation versus gas pressure at the centre of the Sun

At the solar centre, $T_c = 1.57 \times 10^{7}$ K and $P_{\rm gas} \approx 2.3 \times 10^{16}$ Pa. Find $P_{\rm rad}$.

Solution. In the interior the radiation is in TE, so it is isotropic and $P_{\rm rad} = u/3 = aT^4/3$, with $a = 4\sigma/c = 7.566 \times 10^{-16}$ J m^{-3} K^{-4}. Then $u = 4.6 \times 10^{13}$ J m^{-3} and $P_{\rm rad} = 1.5 \times 10^{13}$ Pa, so $P_{\rm rad}/P_{\rm gas} \approx 7 \times 10^{-4}$. Radiation pressure is negligible in the Sun but dominates in very massive stars ($M \gtrsim 50\,M_\odot$).

Worked example 2.3 — Moments of a linear (Eddington) intensity field

Suppose $I(\mu) = I_0 + I_1 \mu$ for all $-1 \le \mu \le 1$. Compute J, H, K and show that $K = J/3$ exactly.

$$J = \tfrac{1}{2}\int_{-1}^{1} (I_0 + I_1\mu)\, d\mu = I_0, \qquad H = \tfrac{1}{2}\int_{-1}^{1} (I_0 + I_1\mu)\mu\, d\mu = \frac{I_1}{3}, \qquad K = \tfrac{1}{2}\int_{-1}^{1} (I_0 + I_1\mu)\mu^2 d\mu =$$

So $K = J/3$ even though the field is **anisotropic** and carries a flux $F = 4\pi H = 4\pi I_1/3$. This is why the Eddington approximation works well deep in a star, where I is very nearly linear in μ.

Exam questions and answers — Lecture 2

Q1. Define I_ν, J_ν, F_ν and K_ν, give their units, and say which can be measured by a single pixel of a telescope pointed at a resolved source.

> **A1.** $I_\nu = dE/(\cos\theta\, dA\, d\Omega\, d\nu\, dt)$ [W m^{-2} Hz^{-1} sr^{-1}]; $J_\nu = \frac{1}{4\pi}\oint I_\nu d\Omega$ [same units]; $F_\nu = \oint I_\nu \cos\theta\, d\Omega$ [W m^{-2} Hz^{-1}]; $K_\nu = \frac{1}{4\pi}\oint I_\nu \cos^2\theta\, d\Omega$ [same as I_ν]. A pixel collects photons from a small solid angle, so it measures I_ν (the surface brightness). For an **unresolved** source the whole object falls in one pixel, and we measure F_ν.

Q2. Prove that I_ν is constant along a ray in vacuum. Hence explain why the Sun's disk looks equally bright from Earth and from Mars.

A2. See §2.3: energy conservation between two areas dA_1, dA_2 at distance r gives $I_1 dA_1 (dA_2/r^2) = I_2 dA_2 (dA_1/r^2)$, so $I_1 = I_2$. From Mars the flux is lower by $(1/1.52)^2 = 0.43$, but the Sun's solid angle is smaller by the same factor. The surface brightness, i.e. I_ν, is the same.

Q3. A surface emits $I(\mu) = I_0$ for $\mu > 0$ and nothing for $\mu < 0$. Find J, F and K at the surface. What is J/F?

A3. $J = \frac{1}{2}\int_0^1 I_0 d\mu = I_0/2$; $F = 2\pi \int_0^1 I_0 \mu \, d\mu = \pi I_0$; $K = \frac{1}{2}\int_0^1 I_0 \mu^2 d\mu = I_0/6$. So $J/F = 1/(2\pi)$, i.e. $J = 2H$ where $H = F/4\pi$. This is **Eddington's surface boundary condition** $J(0) = 2H(0)$, used in Lecture 9.

Q4. A star of radius R and uniform surface intensity I is observed from distance $d \gg R$. Derive $F = \pi I (R/d)^2$, and hence $L = 4\pi R^2 \pi I$.

A4. $F = \int I \cos\theta \, d\Omega$ over the cone $\theta \le \theta_c$, $\sin\theta_c = R/d$: $F = 2\pi I \int_0^{\theta_c} \cos\theta \sin\theta \, d\theta = \pi I \sin^2\theta_c = \pi I R^2/d^2$. Then $L = 4\pi d^2 F = 4\pi R^2 (\pi I)$, consistent with a surface flux πI.

Lecture 3 — Black-Body Radiation: the Planck Function

> **Sources:** MIT 8.901 Lec. 7 (Stefan–Boltzmann, Wien); Hu Set 1; Collins Ch. 9.1 (STE and photon statistics); Rybicki & Lightman Ch. 1 (standard reference).

Learning goals

1. Derive the Planck function from photon statistics.
2. Obtain its Rayleigh–Jeans and Wien limits, Wien's displacement law and the Stefan–Boltzmann law.
3. Define and use brightness, colour and effective temperatures.

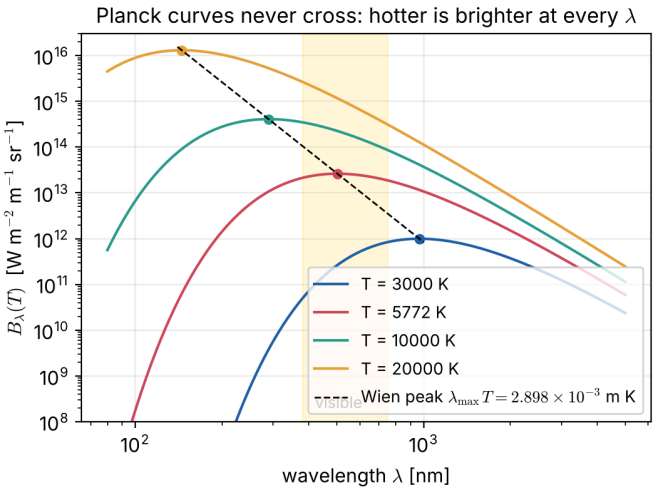
3.1 Black-body radiation

A **black body** is matter that absorbs all radiation falling on it. Radiation in an opaque cavity in thermodynamic equilibrium at temperature T (**Hohlraum** radiation) is isotropic, unpolarised and depends **only on** T. Its intensity is the **Planck function**

$$\boxed{B_\nu(T) = \frac{2h\nu^3}{c^2}\,\frac{1}{e^{h\nu/kT}-1}} \qquad \boxed{B_\lambda(T) = \frac{2hc^2}{\lambda^5}\,\frac{1}{e^{hc/\lambda kT}-1}}$$

Derivation (photon gas)

1. **Number of modes.** In a box of volume V the number of photon states with momentum in d^3p is $2\,V\,d^3p/h^3$ (the factor 2 counts polarisations). With $p = h\nu/c$ and $d^3p = 4\pi p^2 dp$, the number of modes per unit volume in $d\nu$ is $\dfrac{8\pi\nu^2}{c^3}d\nu$.
2. **Mean occupation number.** Photons are bosons with zero chemical potential: $\bar{n} = 1/(e^{h\nu/kT}-1)$.
3. **Energy density.** $u_\nu = \dfrac{8\pi\nu^2}{c^3}\,h\nu\,\bar{n} = \dfrac{8\pi h\nu^3}{c^3}\,\dfrac{1}{e^{h\nu/kT}-1}$.
4. **Intensity.** For isotropic radiation, $u_\nu = 4\pi I_\nu/c$ (Lecture 2), so $I_\nu = cu_\nu/4\pi = B_\nu$. ∎



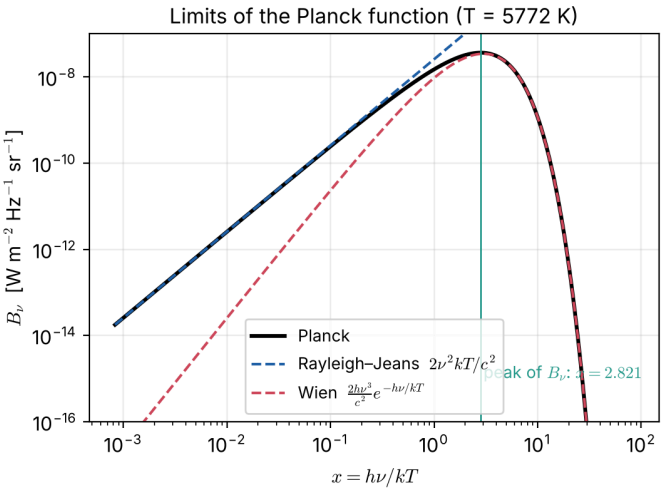


*Figure 3.1 — Left: B_λ for four temperatures, with the Wien peak (dots and dashed line). Hotter bodies are brighter at **every** wavelength, so Planck curves never cross. Right: the Rayleigh–Jeans ($h\nu \ll kT$) and Wien ($h\nu \gg kT$) limits.*

3.2 Limits

Rayleigh–Jeans ($x \equiv h\nu/kT \ll 1$, radio): $e^x - 1 \approx x$, so

$$B_\nu \approx \frac{2\nu^2 kT}{c^2} = \frac{2kT}{\lambda^2}.$$

This is linear in T and contains no h (classical). Integrated over all ν it diverges: the "ultraviolet catastrophe".

Wien ($x \gg 1$, UV and X-ray tail): $B_\nu \approx \dfrac{2h\nu^3}{c^2} e^{-h\nu/kT}$. The steep exponential makes the UV very sensitive to T.

3.3 Wien's displacement law

Set $\partial B_\nu/\partial\nu = 0$: with $x = h\nu/kT$ we need $3(1 - e^{-x}) = x$, whose root is $x = 2.821$:

$$\nu_{\max} = 2.821\frac{kT}{h} = 5.879 \times 10^{10}\ \mathrm{Hz\,K^{-1}} \times T.$$

Set $\partial B_\lambda/\partial\lambda = 0$: $5(1 - e^{-y}) = y$ with $y = hc/\lambda kT$, so $y = 4.965$:

$$\lambda_{\max} T = 2.898 \times 10^{-3}\ \mathrm{m\,K}.$$

Note that $\lambda_{\max} \neq c/\nu_{\max}$: the peak depends on the variable used, because $I_\lambda = (c/\lambda^2) I_\nu$ (Lecture 2, §2.2).

3.4 Stefan–Boltzmann law

$$\int_0^\infty B_\nu\, d\nu = \frac{2h}{c^2}\left(\frac{kT}{h}\right)^4 \int_0^\infty \frac{x^3 dx}{e^x - 1} = \frac{2k^4T^4}{c^2h^3} \cdot \frac{\pi^4}{15} \equiv \frac{\sigma}{\pi} T^4,$$

with

$$\sigma = \frac{2\pi^5 k^4}{15 c^2 h^3} = 5.670 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}.$$

Quantity	Black-body value
Intensity $B = \int B_\nu d\nu$	$\sigma T^4/\pi$
Flux from a surface $F = \pi B$	σT^4
Energy density $u = 4\pi B/c$	aT^4, $a = 4\sigma/c = 7.566 \times 10^{-16}$ J m^{-3} K^{-4}
Pressure $P = u/3$	$aT^4/3$
Photon number density	$n_\gamma = 16\pi\zeta(3)(kT/hc)^3 = 2.03 \times 10^7\, T^3$ m^{-3}
Mean photon energy	$\langle h\nu \rangle = 2.70\, kT$

A useful derivative, needed for the diffusion approximation (Lecture 8):

$$\frac{dB}{dT} = \frac{4\sigma T^3}{\pi}.$$

3.5 Characteristic temperatures

Name	Definition	Use
Brightness temperature $T_b(\nu)$	$I_\nu = B_\nu(T_b)$; in RJ: $T_b = c^2 I_\nu/(2k\nu^2)$	radio astronomy; T_b = actual T only for an optically thick thermal source
Colour temperature T_c	shape (slope) of the spectrum fitted by $B_\nu(T_c)$	photometric colours
Effective temperature $T_{\rm eff}$	$F = \sigma T_{\rm eff}^4$, i.e. $L = 4\pi R^2 \sigma T_{\rm eff}^4$	stars; defines the surface flux
Excitation temperature $T_{\rm ex}$	$n_2/n_1 = (g_2/g_1)e^{-h\nu/kT_{\rm ex}}$	lines; $T_{\rm ex} = T$ only in LTE

For a real star these all differ. They coincide only for a true black body.

Worked example 3.1 — The cosmic microwave background

$T = 2.725$ K. Find (a) $\nu_{\max}$ of B_ν, (b) $\lambda_{\max}$ of B_λ, (c) the photon number and energy densities.

(a) $\nu_{\max} = 5.879\times10^{10}\times 2.725 = 1.60\times10^{11}$ Hz = 160 GHz. **(b)** $\lambda_{\max} = 2.898\times10^{-3}/2.725 = 1.06$ mm. Note that $c/\nu_{\max} = 1.87$ mm $\neq \lambda_{\max}$. **(c)** $n_\gamma = 2.03\times10^{7}\times 2.725^3 = 4.1\times10^{8}$ m^{-3} = **411 photons cm^{-3}**; $u = aT^4 = 4.17\times10^{-14}$ J m^{-3}.

Worked example 3.2 — The Sun at 500 nm

Compute $B_\lambda(500\ \text{nm}, 5772\ \text{K})$ and decide whether the Wien approximation is adequate.

$x = hc/(\lambda kT) = 1.4388\times10^{-2}/(5\times10^{-7}\times 5772) = 4.985$, and $2hc^2/\lambda^5 = 3.811\times10^{15}$ W m^{-3} sr^{-1}, so

$$B_\lambda = \frac{3.811\times10^{15}}{e^{4.985}-1} = 2.62\times10^{13}\ \text{W m}^{-3}\,\text{sr}^{-1}.$$

Wien drops the "-1": the relative error is $e^{-x} = 0.7\,\%$, so Wien is adequate in the visible for solar-type stars.

Worked example 3.3 — Brightness temperature of a radio source

At 1.4 GHz a source has $I_\nu = 2.0\times10^{-19}$ W m^{-2} Hz^{-1} sr^{-1}. Find T_b.

$h\nu/k = 0.067$ K, so the RJ limit is excellent for any $T \gg 0.07$ K:

$$T_b = \frac{c^2 I_\nu}{2k\nu^2} = \frac{(3\times10^8)^2\times 2\times10^{-19}}{2\times1.381\times10^{-23}\times(1.4\times10^9)^2} = 332\ \text{K}.$$

If the source is an optically thin H II region with $T_e = 8000$ K, then $T_b = T_e\tau$ (Lecture 5), so $\tau \approx 0.04$.

Worked example 3.4 — Inverting the Planck function

An observed intensity equals B_λ; solve for T.

$$T = \frac{hc}{\lambda k}\,\frac{1}{\ln\!\left(1 + \dfrac{2hc^2}{\lambda^5 B_\lambda}\right)}.$$

For $\lambda = 500$ nm and $B_\lambda = 4.0 \times 10^{13}$ W m^{-3} sr^{-1}: $T = 28\,776/\ln(1 + 95.3) = 6300$ K. We use this formula in Lecture 7 to map out the solar temperature from limb darkening.

Exam questions and answers — Lecture 3

Q1. Derive the Planck function from the density of photon states and Bose–Einstein statistics.

> **A1.** See §3.1. Modes per unit volume per $d\nu$: $8\pi\nu^2/c^3$. Mean energy per mode: $h\nu/(e^{h\nu/kT} - 1)$. Therefore $u_\nu = (8\pi h\nu^3/c^3)/(e^{h\nu/kT} - 1)$, and $B_\nu = cu_\nu/4\pi$.

Q2. Show that B_ν has its maximum at $h\nu = 2.821\,kT$. Is the peak of B_λ at $\lambda = c/\nu_{\max}$?

> **A2.** $B_\nu \propto x^3/(e^x - 1)$; $d/dx = 0 \Rightarrow 3x^2(e^x - 1) - x^3 e^x = 0 \Rightarrow 3(1 - e^{-x}) = x$, whose root is $x = 2.821$ (solve by iteration: $x_{n+1} = 3(1 - e^{-x_n})$). For $B_\lambda \propto y^5/(e^y - 1)$ the condition is $5(1 - e^{-y}) = y$, $y = 4.965$. Since $4.965 \neq 2.821$, $\lambda_{\max} \neq c/\nu_{\max}$; the two differ by a factor 1.76.

Q3. Derive the Stefan–Boltzmann law, giving σ in terms of fundamental constants. (Use $\int_0^\infty x^3/(e^x - 1)\,dx = \pi^4/15$.)

> **A3.** $F = \pi \int B_\nu d\nu = \pi \frac{2h}{c^2}\left(\frac{kT}{h}\right)^4 \frac{\pi^4}{15} = \frac{2\pi^5 k^4}{15 c^2 h^3} T^4 = \sigma T^4$.

Q4. Why is T_b of an optically thin source **less** than its kinetic temperature? Under what condition are they equal?

> **A4.** For an isothermal slab with no background, $I_\nu = B_\nu(T)(1 - e^{-\tau_\nu})$ (Lecture 5), so in RJ $T_b = T(1 - e^{-\tau}) < T$, and $T_b \approx T\tau$ when $\tau \ll 1$. They are equal only when $\tau_\nu \gg 1$ (and the source is thermal, $S_\nu = B_\nu$).

Lecture 4 — Emission, Absorption and Scattering

> **Sources:** Hu Set 1 (spontaneous emission, absorption, cross-section, scattering, random walk, multiple processes); MIT 8.901 Lec. 7 ($\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$); Collins §9.2c (creation rate, redistribution function); Chandrasekhar Ch. I §§7–11.

Learning goals

1. Define j_ν, α_ν, κ_ν, σ_ν and the mean free path.
2. Derive the Einstein relations and the correction for stimulated emission.
3. Describe scattering as absorption followed by re-emission, and write the mixed source function.
4. Use the random walk to estimate photon escape times.

4.1 The four radiative processes

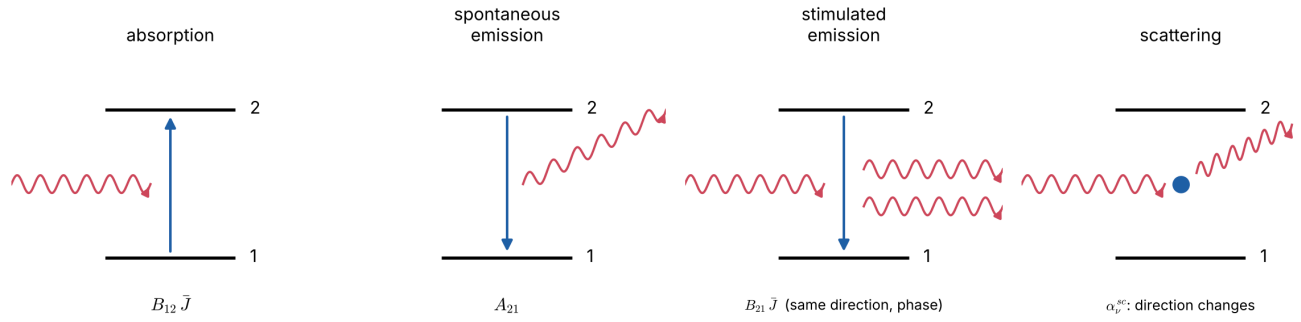


*Figure 4.1 — Absorption, spontaneous emission, stimulated emission and scattering. Stimulated emission produces a photon identical to the incoming one (same frequency, direction and polarisation), so mathematically it behaves like **negative absorption**.*

4.2 Emission coefficient

Let matter emit power $P_\nu\, d\nu$ per unit volume. If it emits isotropically, the energy into $d\Omega$ is

$$dE = j_\nu\, dV\, d\Omega\, d\nu\, dt, \qquad j_\nu = \frac{P_\nu}{4\pi}.$$

Along a path ds through a cylinder of cross-section dA ($dV = dA\, ds$), the beam gains $dI_\nu = j_\nu\, ds$. With emission only:

$$\frac{dI_\nu}{ds} = j_\nu \quad \Rightarrow \quad I_\nu(s) = I_\nu(s_0) + \int_{s_0}^{s} j_\nu\, ds'.$$

The emissivity per unit mass is $\epsilon_\nu = 4\pi j_\nu / \rho$ (Hu).

4.3 Absorption coefficient, cross-section and opacity

A beam loses a fraction $\alpha_\nu\, ds$ of its intensity over ds:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu \quad \Rightarrow \quad I_\nu(s) = I_\nu(0)\, e^{-\int \alpha_\nu ds} = I_\nu(0) e^{-\tau_\nu}.$$

Particle picture (Hu). Absorbers of number density n each present a cross-section σ_ν. In a slab $dA \times ds$ they block an area $\sigma_\nu n\, dA\, ds$, so the fraction removed is $n\sigma_\nu ds$:

$$\boxed{\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu} \qquad \text{mean free path } \ell_\nu = \frac{1}{\alpha_\nu}.$$

An optical depth τ_ν therefore equals the number of mean free paths along the path.

Main opacity sources in stars

Process	Name	Frequency dependence	Where it matters
bound–bound	line absorption	narrow profile $\phi(\nu)$	spectral lines
bound–free	photo-ionisation	edges, $\propto \nu^{-3}$ above each edge	Balmer/Lyman jumps; H^- in the Sun
free–free	bremsstrahlung	$\propto \nu^{-3}$ (optical), ν^{-2} (radio)	hot stars, H II regions
electron scattering	Thomson	grey: $\sigma_T = 6.652 \times 10^{-29}$ m^2	O stars, interiors, accretion disks
H^-	b-f + f-f of the negative H ion	broad	**dominant continuum opacity in the solar photosphere**

For fully ionised gas, $\kappa_{\rm es} = \sigma_T/(\mu_e m_H) = 0.020(1+X)$ m^2 kg^{-1} $= 0.2(1+X)$ cm^2 g^{-1}.

4.4 Einstein coefficients

Consider a transition $1 \leftrightarrow 2$ with energy $h\nu_0 = E_2 - E_1$, statistical weights g_1, g_2, and line profile $\phi(\nu)$ normalised to $\int \phi\, d\nu = 1$. Let $\bar{J} = \int J_\nu \phi(\nu)\, d\nu$.

Process	Rate per unit volume
spontaneous emission	$n_2 A_{21}$
absorption	$n_1 B_{12} \bar{J}$
stimulated emission	$n_2 B_{21} \bar{J}$

Relations. In TE, $J_\nu = B_\nu(T)$, detailed balance holds ($n_1 B_{12}\bar{J} = n_2 A_{21} + n_2 B_{21}\bar{J}$), and the populations follow Boltzmann, $n_2/n_1 = (g_2/g_1) e^{-h\nu/kT}$. Solving for $\bar{J}$:

$$\bar{J} = \frac{A_{21}/B_{21}}{(g_1 B_{12}/g_2 B_{21}) e^{h\nu/kT} - 1}.$$

For this to equal $B_\nu(T)$ at **every** T we need

$$\boxed{g_1 B_{12} = g_2 B_{21}, \qquad A_{21} = \frac{2h\nu^3}{c^2} B_{21}}$$

These are **atomic** properties, so they hold whether or not the gas is in equilibrium.

Macroscopic coefficients. Each transition moves energy $h\nu$, spread over 4π sr and over the profile:

$$j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi(\nu), \qquad \alpha_\nu = \frac{h\nu}{4\pi}\left(n_1 B_{12} - n_2 B_{21}\right)\phi(\nu) = \frac{h\nu}{4\pi} n_1 B_{12}\left(1 - \frac{g_1 n_2}{g_2 n_1}\right)\phi(\nu).$$

The bracket is the **correction for stimulated emission**. If $g_1 n_2 / g_2 n_1 > 1$ (a population inversion), then $\alpha_\nu < 0$ and the beam grows exponentially: a **maser**.

Line source function (any populations):

$$S_\nu = \frac{j_\nu}{\alpha_\nu} = \frac{2h\nu^3}{c^2}\,\frac{1}{\dfrac{g_2 n_1}{g_1 n_2} - 1}.$$

With Boltzmann populations at T this becomes exactly $B_\nu(T)$. This is the key LTE result of Lecture 6.

4.5 Scattering

Scattering is absorption immediately followed by re-emission into a new direction (Hu). For **coherent** (same frequency), **isotropic** scattering with coefficient $\alpha_\nu^{\rm sc}$, the energy removed from all directions is redistributed evenly over 4π:

$$j_\nu^{\rm sc} = \alpha_\nu^{\rm sc} \oint I_\nu \frac{d\Omega}{4\pi} = \alpha_\nu^{\rm sc} J_\nu \quad \Rightarrow \quad S_\nu^{\rm sc} = J_\nu.$$

The RTE then becomes an **integro-differential equation**, because J_ν depends on I_ν in all directions:

$$\frac{dI_\nu}{ds} = -\alpha_\nu^{\rm sc}\left(I_\nu - \oint I_\nu \frac{d\Omega}{4\pi}\right).$$

Collins (eq. 9.2.27) gives the general form with a redistribution function $R(\nu', \nu, \Omega', \Omega)$, describing how photons are moved in frequency and angle.

Absorption plus scattering

The coefficients add (Hu): $\alpha_\nu = \alpha_\nu^a + \alpha_\nu^{\rm sc}$. The combined source function is

$$S_\nu = \frac{\alpha_\nu^a B_\nu + \alpha_\nu^{\rm sc} J_\nu}{\alpha_\nu^a + \alpha_\nu^{\rm sc}} = \varepsilon_\nu B_\nu + (1-\varepsilon_\nu) J_\nu, \qquad \varepsilon_\nu = \frac{\alpha_\nu^a}{\alpha_\nu^a + \alpha_\nu^{\rm sc}},$$

where $1 - \varepsilon_\nu$ is the **single-scattering albedo** (the probability that an interaction is a scattering). This equation is the heart of Lecture 8.

4.6 Random walk

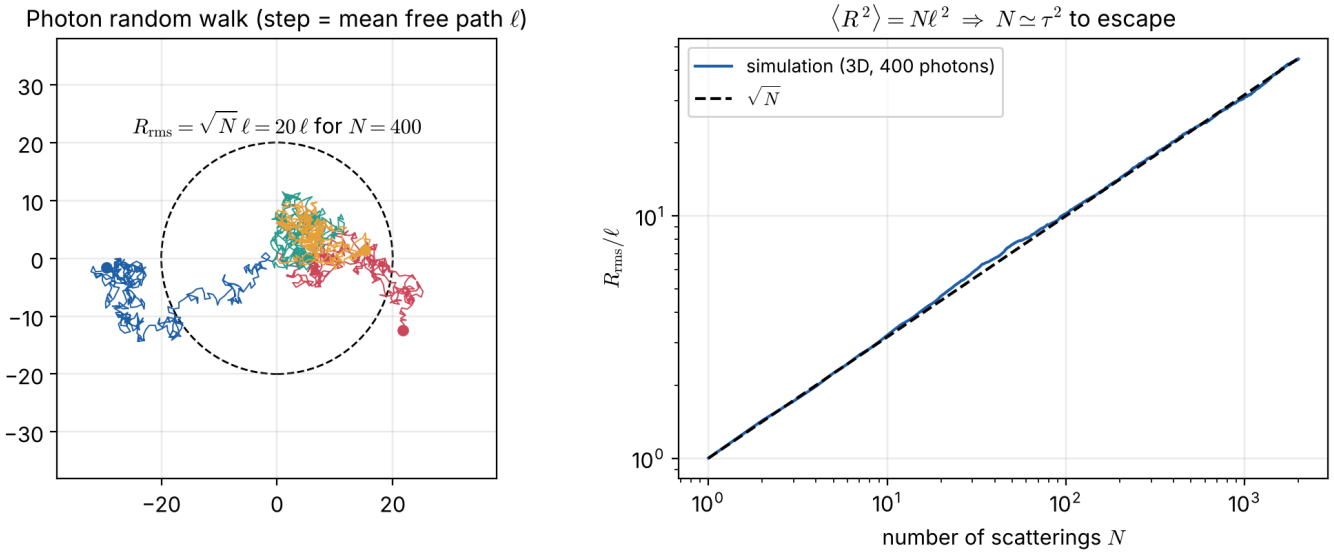


Figure 4.2 — Left: photon paths with unit steps; the dashed circle has radius $\sqrt{N}\ell$. Right: a 3D simulation confirms $R_{\rm rms} = \sqrt{N}\,\ell$.

With steps $\mathbf{r}_i$ of length ℓ in random directions, $\mathbf{R} = \sum \mathbf{r}_i$ and $\langle \mathbf{r}_i \cdot \mathbf{r}_j \rangle = \ell^2 \delta_{ij}$, so

$$\langle R^2 \rangle = N\ell^2 .$$

To escape from optical depth $\tau = R/\ell$ takes $N \approx \tau^2$ steps when $\tau \gg 1$. When $\tau \ll 1$, only a fraction τ of photons interact at all. A useful estimate is $N \approx \max(\tau, \tau^2)$ (Hu).

Thermalisation. If each interaction is an absorption with probability ε, a photon survives $N \approx 1/\varepsilon$ interactions before it is destroyed, travelling a net optical distance $\tau_{\rm th} \approx \sqrt{N} = 1/\sqrt{\varepsilon}$: the **thermalisation depth** (Lecture 8).

Worked example 4.1 — How long does light take to leave the Sun?

Assume a uniform Sun of mean density $\bar\rho$, fully ionised with $X = 0.7$, and only Thomson scattering.

1. $\bar\rho = M_\odot / (\tfrac{4}{3}\pi R_\odot^3) = 1410$ kg m^{-3}.
2. $\mu_e = 2/(1+X) = 1.18$, so $n_e = \bar\rho/(\mu_e m_H) = 7.2 \times 10^{29}$ m^{-3}.
3. $\ell = 1/(n_e \sigma_T) = 2.1$ cm.
4. $\tau = R_\odot/\ell = 3.3 \times 10^{10}$, so $N = \tau^2 = 1.1 \times 10^{21}$ scatterings.
5. Total path $N\ell = R^2/\ell$; time $t = R_\odot^2/(\ell c) = 7.7 \times 10^{10}$ s $\approx$ **2400 yr**, compared with a light-crossing time of 2.3 s.

The real Sun is denser in the core and has higher (bound-free) opacity, so realistic estimates are 10^4–10^5 yr. The method is the point.

Worked example 4.2 — Stimulated emission at radio wavelengths

Evaluate $1 - e^{-h\nu/kT}$ (the LTE correction factor) for the 21-cm line at $T = 100$ K and for Hα in the Sun.

21 cm: $h\nu/kT = 6.8 \times 10^{-4}$, so the factor is 6.8×10^{-4}. Stimulated emission cancels **99.93 %** of the absorption, and the net opacity $\propto 1/T$. That is why 21-cm absorption measures the spin temperature. Hα (656 nm, 5772 K): $h\nu/kT = 3.8$, so the factor is 0.98; stimulated emission is a 2 % correction.

Worked example 4.3 — Source function of a two-level atom

Show that if the level populations are Boltzmann at temperature $T_{\rm ex}$, then $S_\nu = B_\nu(T_{\rm ex})$.

Insert $g_2 n_1/(g_1 n_2) = e^{h\nu/kT_{\rm ex}}$ into $S_\nu = (2h\nu^3/c^2)/[g_2 n_1/(g_1 n_2) - 1]$ to get $S_\nu = B_\nu(T_{\rm ex})$. **The line source function only measures the excitation temperature.** LTE is the special case $T_{\rm ex} = T_{\rm kin}$.

Exam questions and answers — Lecture 4

Q1. Derive the Einstein relations. Explain why they hold even when the gas is not in TE.

> **A1.** See §4.4. Detailed balance plus Boltzmann populations, required to reproduce $B_\nu(T)$ for all T, gives $g_1 B_{12} = g_2 B_{21}$ (high-T limit) and $A_{21} = (2h\nu^3/c^2) B_{21}$. The coefficients depend only on the atomic wavefunctions (matrix elements), not on the environment, so once found in TE they apply everywhere.

Q2. Write α_ν including stimulated emission. What happens physically when $n_2/g_2 > n_1/g_1$? Give an astrophysical example.

> **A2.** $\alpha_\nu = \frac{h\nu}{4\pi}\phi(\nu)(n_1 B_{12} - n_2 B_{21}) = \frac{h\nu}{4\pi}\phi\, n_1 B_{12}(1 - \frac{g_1 n_2}{g_2 n_1})$. With an inversion, $\alpha_\nu < 0$ and $\tau_\nu < 0$, so I_ν grows as $e^{|\tau|}$: maser amplification. Examples: OH (1665/1667 MHz) and H_2O (22 GHz) masers in star-forming regions and the envelopes of AGB stars.

Q3. Derive $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$ and give the meaning of ε.

> **A3.** $j = \alpha^a B + \alpha^{\rm sc} J$ (thermal emission by Kirchhoff, plus scattering) and $\alpha = \alpha^a + \alpha^{\rm sc}$. Divide to get $S = \varepsilon B + (1-\varepsilon) J$ with $\varepsilon = \alpha^a/(\alpha^a + \alpha^{\rm sc})$, the probability that an interaction destroys the photon (converts it into thermal energy of the gas).

Q4. A cloud has $\tau = 100$ for pure scattering. Roughly how many scatterings does a photon undergo before escaping, and how far does it travel in total?

> **A4.** $N \approx \tau^2 = 10^4$; total path $N\ell = \tau^2 \ell = 100\, L$, where $L = \tau\ell$ is the size of the cloud. So the photon travels 100 times the cloud size, and the escape time is $100\, L/c$.

Lecture 5 — The Radiative Transfer Equation and Its Formal Solution

> **Sources:** MIT 8.901 Lec. 7 §§7.1–7.3 (RTE built up in three steps, constant-S solution, formal solution, Kirchhoff's law for thick and thin lumps, thin H cloud, two-layer stellar atmosphere, thermal inversions); Hu Set 1 (optical depth, source function, formal solution, interpretation); Collins §9.2c–d (source function and its meaning).

Learning goals

1. Write the RTE in terms of τ_ν and S_ν and explain why "I_ν relaxes toward S_ν".
2. Derive the formal solution with an integrating factor.
3. Solve the uniform slab, including its optically thin and thick limits.
4. Explain Kirchhoff's three laws of spectroscopy with RT, including MIT's two-layer star and thermal inversions.

5.1 The equation, built up in three steps (MIT 8.901 §7.1)

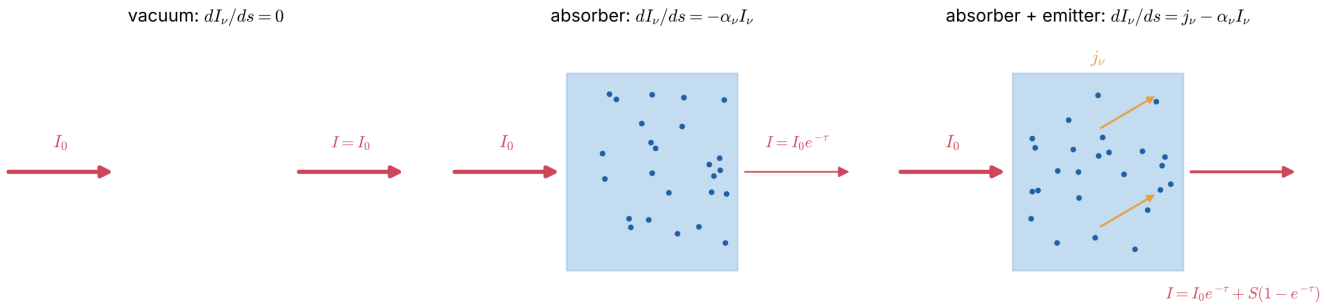


Figure 5.0 — After MIT 8.901 Lecture 7, Fig. 9: radiation through (left) vacuum, $dI_\nu/ds = 0$; (centre) a purely absorbing medium, $dI_\nu/ds = -\alpha_\nu I_\nu$, so $I = I_0 e^{-\tau}$; (right) a medium that absorbs and emits, $dI_\nu/ds = j_\nu - \alpha_\nu I_\nu$.

MIT's notation: $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu$ (extinction coefficient, m^{-1}) and j_ν (emissivity, $\mathrm{W\,m^{-3}\,Hz^{-1}\,sr^{-1}}$). Note that j_ν and I_ν have different units; their ratio $S_\nu = j_\nu/\alpha_\nu$ has the units of intensity. Combining Lecture 4's gains and losses:

$$\frac{dI_\nu}{ds} = -\alpha_\nu I_\nu + j_\nu.$$

Divide by α_ν and use $d\tau_\nu = \alpha_\nu\, ds$ (optical depth measured **along the beam**) and $S_\nu = j_\nu/\alpha_\nu$:

$$\boxed{\frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu}$$

Physical meaning. If $I_\nu > S_\nu$, then $dI_\nu/d\tau_\nu < 0$: intensity decreases. If $I_\nu < S_\nu$, it increases. **The intensity always relaxes toward the local source function, on a scale of one optical depth.** S_ν is the intensity the medium

"wants" to have.

5.2 Formal solution

Multiply by the integrating factor e^{τ_ν} (MIT eq. 82–85):

$$\frac{d}{d\tau_\nu}\left(I_\nu e^{\tau_\nu}\right) = S_\nu e^{\tau_\nu}.$$

Integrate from 0 to τ_ν:

$$\boxed{I_\nu(\tau_\nu) = I_\nu(0)\, e^{-\tau_\nu} + \int_0^{\tau_\nu} S_\nu(t)\, e^{-(\tau_\nu - t)}\, dt}$$

Interpretation (Hu): what arrives at τ_ν is (i) the initial intensity attenuated by the whole column, plus (ii) the emission $S_\nu\, dt$ from every layer t, attenuated by the material between t and the observer, $e^{-(\tau_\nu - t)}$.

The solution is "formal" because S_ν may depend on I_ν itself (scattering, NLTE). It is then an integral equation, not an explicit answer. In **LTE without scattering**, $S_\nu = B_\nu(T)$ is known once $T(\tau)$ is known, and the formal solution **is** the solution.

5.3 The uniform (isothermal) slab

If S_ν is constant:

$$\boxed{I_\nu = I_\nu(0)\, e^{-\tau_\nu} + S_\nu\left(1 - e^{-\tau_\nu}\right)}$$

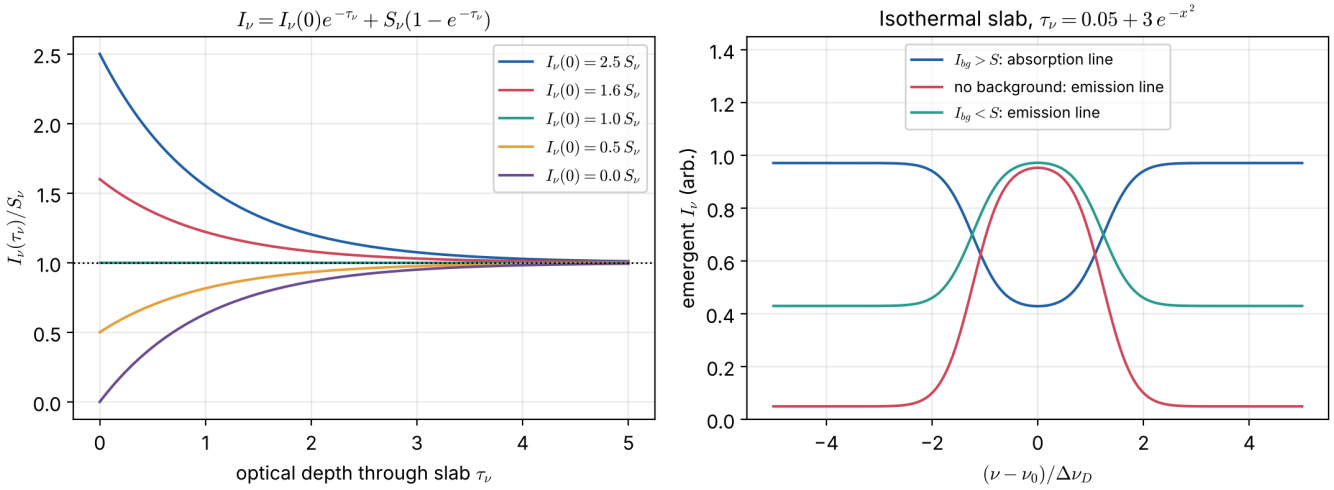


Figure 5.1 — Left: whatever the incident intensity, $I_\nu \to S_\nu$ after a few optical depths. Right: a slab with optical depth larger at line centre than in the wings produces an absorption line when the background is brighter than S, and an emission line when it is fainter.

Limit	Result	Meaning
$\tau_\nu \ll 1$ (optically thin)	$I_\nu \approx I_\nu(0)(1-\tau_\nu) + \tau_\nu S_\nu$	background passes; emission $\propto$ column density ($\tau S = j\,L$)
$\tau_\nu \gg 1$ (optically thick)	$I_\nu \to S_\nu$	memory of the background lost; we see the medium itself
thick, LTE	$I_\nu = B_\nu(T)$	an optically thick isothermal LTE body radiates as a black body

> Even in LTE, $I_\nu = B_\nu$ only inside an **optically thick, isothermal** body. A real star is not isothermal, so its emergent spectrum is not exactly Planckian.

Kirchhoff's laws of spectroscopy, explained

Write $\tau_\nu = \tau_c + \tau_0\phi(\nu)$, with a weak continuum τ_c plus a line. Then:

1. **Hot, opaque body** ($\tau_\nu \gg 1$ at all ν): $I_\nu = B_\nu(T)$, a continuous spectrum.
2. **Hot, thin gas, no background** ($I_\nu(0) = 0$): $I_\nu = B_\nu(1 - e^{-\tau_\nu})$ is largest where τ_ν is largest, so we see **emission lines**.
3. **Cooler gas in front of a hotter continuum** ($I_\nu(0) > S_\nu$): where τ_ν is large, I_ν is pulled *down* toward S_ν, giving **absorption lines**.

The real criterion is $I_{\text{background}}$ versus S_ν, not "cool versus hot" by itself. This idea carries over to stellar atmospheres in Lecture 10.

5.4 MIT 8.901 §7.3: thin clouds, two-layer stars and thermal inversions

MIT's form of **Kirchhoff's law** says that a thermally emitting object in equilibrium has $S_\nu = B_\nu(T)$, but this does **not** mean that it emits a black-body spectrum. What emerges depends on τ_ν. Take two lumps of matter at the same T, with nothing behind them ($I_\nu(0) = 0$):

Lump	Emergent intensity	Spectrum
optically thick, $\tau_\nu \gg 1$	$I_\nu = S_\nu = B_\nu(T)$	black body
optically thin, $\tau_\nu \ll 1$	$I_\nu = S_\nu\,[1-(1-\tau_\nu)] = \tau_\nu B_\nu(T) = \alpha_\nu s\, B_\nu(T)$	black body × τ_ν: bright only where α_ν is large

(a) A hot, thin hydrogen cloud on a dark sky. α_ν is large only at the hydrogen lines, whose wavelengths follow the Rydberg formula

$$\frac{1}{\lambda_{\text{vac}}} = R_{\text{H}}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \qquad R_{\text{H}} = 1.0968 \times 10^7\ \text{m}^{-1}$$

($n_1 = 1, 2, 3, 4$: Lyman, Balmer, Paschen, Brackett). Since $I_\nu = \tau_\nu B_\nu$, we see an **emission-line spectrum** (Figure 5.2a).

(b) The simplest stellar atmosphere (MIT Fig. 10): an optically thick interior at T_H under a cooler, thin layer at $T_C < T_H$. The thick base supplies $I_\nu(0) = B_\nu(T_H)$, and the layer gives

$$I_\nu = B_\nu(T_H)e^{-\tau_\nu} + B_\nu(T_C)\left(1 - e^{-\tau_\nu}\right) \overset{\tau_\nu \ll 1}{\approx} B_\nu(T_H) - \tau_\nu\left[B_\nu(T_H) - B_\nu(T_C)\right].$$

The **continuum** (small α_ν) shows $B_\nu(T_H)$ from deep down. At the **lines** (large α_ν) we see only down to $\tau_\nu \sim 1$, i.e. the cool layer, so $I_\nu \to B_\nu(T_C)$: **absorption lines** (Figure 5.2b). An isothermal atmosphere gives a pure black body whatever α_ν is.

(c) Thermal inversion. If T *increases* outward ($T_{\rm top} > T_{\rm base}$), every argument reverses and the same lines appear in **emission** above the continuum (Figure 5.2c). MIT notes that inversions are ubiquitous: the solar chromosphere, Solar System planets (stratospheres) and hot-Jupiter exoplanets.

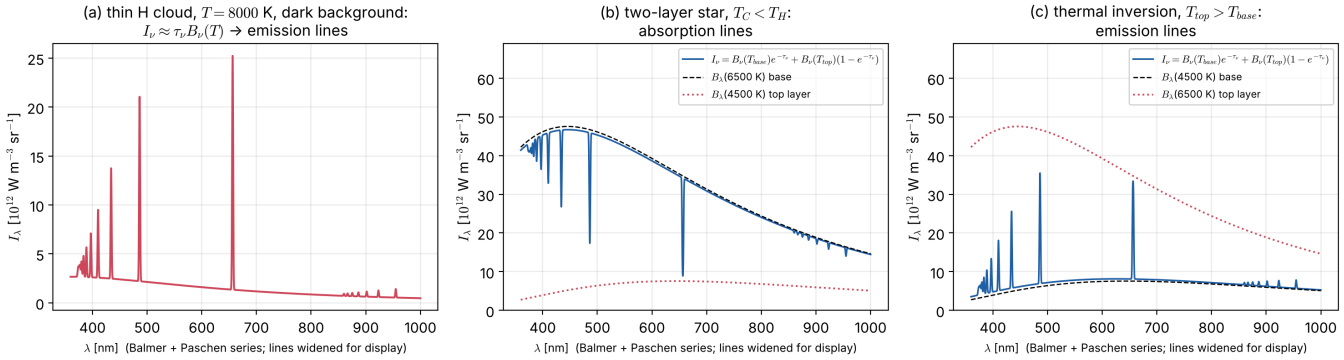


Figure 5.2 — The MIT 8.901 toy models, computed exactly with $I = I_0 e^{-\tau} + B(1 - e^{-\tau})$ for Balmer and Paschen lines placed by the Rydberg formula (line widths exaggerated for visibility). (a) Thin 8000 K cloud: emission lines. (b) 6500 K base under a 4500 K layer: absorption lines. (c) Inverted case: emission lines.

5.5 Radio spectrum of an H II region

For free–free absorption, $\tau_\nu \propto T_e^{-1.35}\nu^{-2.1}\int n_e^2\, ds$ (the integral is the emission measure). In the RJ limit, $I_\nu \propto T_b$:

$$T_b = T_e\left(1 - e^{-\tau_\nu}\right).$$

At low ν, $\tau \gg 1$, so $T_b = T_e$ and the flux $S_\nu \propto \nu^2 T_e$ (black body). At high ν, $\tau \ll 1$, so $T_b = T_e\tau_\nu \propto \nu^{-2.1}$ and the flux is $\propto \nu^{-0.1}$ (almost flat). The turnover at $\tau_\nu \approx 1$ measures the emission measure.

Worked example 5.1 — H II region brightness temperatures

$T_e = 8000$ *K. Find* T_b *for* $\tau = 0.01, 0.1, 1, 3, 10$.

τ	**0.01**	**0.1**	**1**	**3**	**10**
$T_b = T_e(1 - e^{-\tau})$ [K]	80	761	5057	7602	8000

Lesson: the brightness temperature equals the physical temperature only when the source is optically thick.

Worked example 5.2 — Absorption by a foreground cloud

A star with continuum intensity I_0 is seen through a cloud with $S = 0.3 I_0$. At line centre $\tau = 5$, in the wings $\tau = 0.1$. Find the line profile values and the central depth.

$I = I_0 e^{-\tau} + S(1 - e^{-\tau})$:

- wings ($\tau = 0.1$): $I = 0.933\, I_0$;
- $\tau = 1$: $I = 0.558\, I_0$;
- centre ($\tau = 5$): $I = 0.305\, I_0 \approx S$.

The line core is **saturated**: it cannot go below S, however large τ becomes. The central depth is $1 - 0.305/0.933 = 0.67$ relative to the local continuum.

Worked example 5.3 — Formal solution with a linearly increasing source function

Let $S(t) = a + bt$ along a path of total optical depth τ, with $I(0) = 0$. Find $I(\tau)$ and its thick limit.

$$I(\tau) = \int_0^\tau (a + bt) e^{-(\tau - t)} dt.$$

Put $u = \tau - t$: $I = \int_0^\tau [a + b(\tau - u)] e^{-u} du = (a + b\tau)(1 - e^{-\tau}) - b\,[1 - (1 + \tau)e^{-\tau}]$. For $\tau \to \infty$: $I \to a + b\tau - b = S(\tau - 1)$.

The emerging intensity equals the source function one optical depth back from the observer. This is the seed of the Eddington–Barbier relation (Lecture 7).

Worked example 5.4 — The MIT two-layer star at Hα

Base $T_H = 6500$ K, upper layer $T_C = 4500$ K. At $\lambda = 656.3$ nm the layer has $\tau_c = 0.02$ in the continuum and $\tau = 2.02$ at the centre of Hα. (a) Find the line depth. (b) Compare with the linearised MIT formula, and say when the formula fails. (c) What is the deepest the line could ever be?

$B_\lambda(6500\ \mathrm{K}) = 3.474 \times 10^{13}$ and $B_\lambda(4500\ \mathrm{K}) = 7.551 \times 10^{12}$ W m^{-3} sr^{-1}.

(a) Continuum: $I_c = B_H e^{-0.02} + B_C(1 - e^{-0.02}) = 0.9845\, B_H$. Line: $I_l = B_H e^{-2.02} + B_C(1 - e^{-2.02}) = 0.321\, B_H$. Depth $= 1 - I_l/I_c = 0.67$.

(b) The linearised form $B_H - \tau(B_H - B_C)$ gives $0.984\, B_H$ for $\tau = 0.02$ (correct to 0.02 %) but $-0.57\, B_H$ for $\tau = 2$: **negative and meaningless**. The linear form holds only for $\tau_\nu \ll 1$; strong lines need the full exponential.

(c) For $\tau \to \infty$, $I_l \to B_C$, so the maximum depth is $1 - B_C/I_c = 0.78$. **The depth of a saturated line measures the temperature contrast of the atmosphere**, which is the same conclusion as the Milne–Eddington model of Lecture 10.

Exam questions and answers — Lecture 5

Q1. Starting from $dI_\nu/ds = -\alpha_\nu I_\nu + j_\nu$, derive the formal solution.

A1. With $d\tau = \alpha\, ds$ and $S = j/\alpha$: $dI/d\tau + I = S$. Multiply by e^{τ}: $d(Ie^{\tau})/d\tau = Se^{\tau}$. Integrate: $I(\tau)e^{\tau} - I(0) = \int_0^{\tau} S(t)e^{t}dt$, so $I(\tau) = I(0)e^{-\tau} + \int_0^{\tau} S(t)e^{-(\tau - t)}dt$.

Q2. For a uniform slab with S constant, derive $I = I_0 e^{-\tau} + S(1 - e^{-\tau})$ and discuss the thin and thick limits.

A2. Take S out of the integral: $\int_0^{\tau} e^{-(\tau - t)}dt = 1 - e^{-\tau}$. Thin: $I \approx I_0 + \tau(S - I_0)$, so the change is linear in τ (the column density). Thick: $I \to S$, independent of I_0.

Q3. "Absorption lines form because the gas in a stellar atmosphere is cooler than the gas below it." Rephrase this precisely using I_ν and S_ν.

A3. At line frequencies τ_ν is large, so the emergent intensity is set by S_ν in higher layers. If S_ν decreases outward (in LTE, $S_\nu = B_\nu(T)$ with T decreasing outward), then line photons come from layers with smaller S_ν than continuum photons, and an absorption line results. If S_ν increases outward (a chromosphere, or NLTE effects), the line appears in emission.

Q4. (MIT 8.901 two-layer model.) A star has an optically thick base at T_H and a thin upper layer at T_C. Show that $I_\nu \approx B_\nu(T_H) - \tau_\nu[B_\nu(T_H) - B_\nu(T_C)]$ for $\tau_\nu \ll 1$. Explain why the spectrum shows absorption lines, and what changes in a thermal inversion.

A4. The base is thick, so $I_\nu(0) = B_\nu(T_H)$. The layer has $S_\nu = B_\nu(T_C)$ (LTE), so $I_\nu = B_\nu(T_H)e^{-\tau_\nu} + B_\nu(T_C)(1 - e^{-\tau_\nu})$, and expanding $e^{-\tau} \approx 1 - \tau$ gives the result. Where α_ν is large (lines), τ_ν is large and the intensity drops toward $B_\nu(T_C) < B_\nu(T_H)$: absorption lines. In a thermal inversion $T_C > T_H$, so $B_\nu(T_C) > B_\nu(T_H)$ and the same lines appear in emission (solar chromosphere, planetary stratospheres). For strong lines the linear form fails (it can even go negative), and the full exponential must be used (Worked example 5.4).

Lecture 6 — Thermodynamic Equilibrium and Local Thermodynamic Equilibrium (LTE)

> **Sources:** Collins §9.1a–d (breakdown of STE, assumption of LTE, the five assumptions of normal stellar atmospheres), §9.2c (Kirchhoff's law $j_\nu = \kappa_\nu B_\nu$, eq. 9.2.22), §9.2d ($S_\nu = B_\nu$, eq. 9.2.28); MIT 8.901 §7.3 (Kirchhoff's law for thick and thin lumps); Hu Set 1; Carroll & Ostlie Ch. 8 (Saha–Boltzmann example).

Learning goals

1. State what is in equilibrium in TE and in LTE.
2. Derive Kirchhoff's law $j_\nu = \alpha_\nu B_\nu(T)$ and the LTE result $S_\nu = B_\nu(T)$.
3. Use the Boltzmann and Saha equations; explain the strength of the Balmer lines along the spectral sequence.
4. Derive the two-level-atom source function and the condition for LTE to be valid.

6.1 Strict (full) thermodynamic equilibrium

In TE at temperature T, **every** distribution is the equilibrium one:

Distribution	Law
particle velocities	Maxwell: $f(v) \propto v^2 e^{-mv^2/2kT}$
excitation	Boltzmann: $\dfrac{n_j}{n_i} = \dfrac{g_j}{g_i} e^{-(E_j - E_i)/kT}$
ionisation	Saha (below)
radiation	Planck: $I_\nu = B_\nu(T)$, isotropic

Deep in a star this holds extremely well. The photon mean free path ($\sim$ cm) is tiny compared with the temperature scale height ($\sim R_\odot \sim 10^{11}$ cm), so every photon sees an isothermal enclosure.

6.2 Why TE fails near the surface

Collins (§9.1a) identifies the key asymmetry: **photons have much longer mean free paths than particles**. Approaching the surface:

1. Photons begin to **escape** into space, so I_ν becomes anisotropic and departs from B_ν. The radiation field "feels" the boundary first.
2. A photon at a given point was emitted in regions with different T; the radiation field is **non-local**.
3. Particles, with short mean free paths, still collide often enough to keep Maxwell, Boltzmann and Saha distributions at the **local** T.

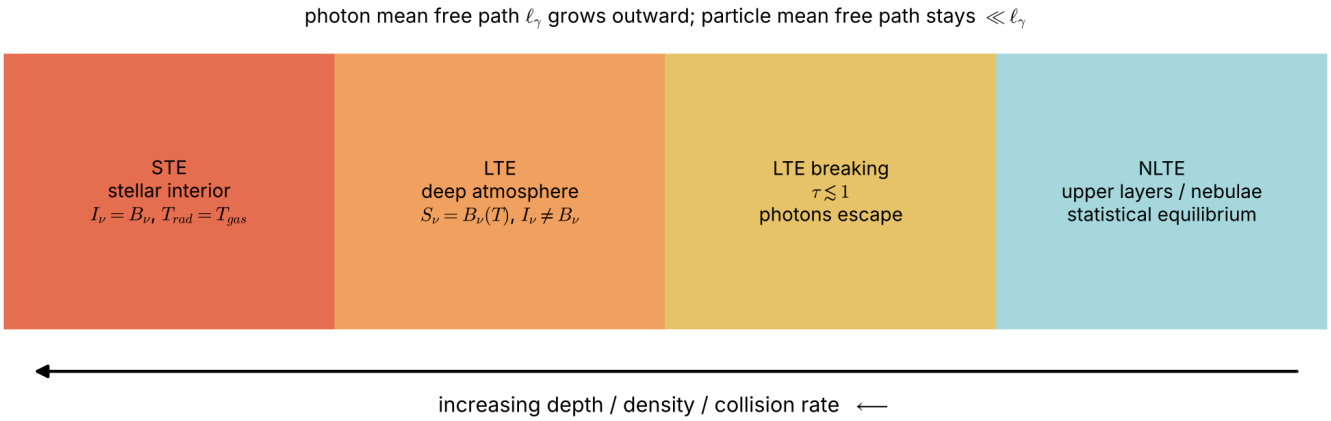


Figure 6.1 — From STE in the interior, through LTE in the deep atmosphere, to NLTE in the upper layers. LTE fails first for strong lines and at low density.

6.3 Local thermodynamic equilibrium

> **Definition.** In **LTE** the *matter* at each point is in thermodynamic equilibrium at the local temperature $T(\mathbf{r})$: velocities are Maxwellian, populations follow Boltzmann and Saha. The radiation field I_ν is **not** required to be Planckian.

Kirchhoff's law

In a closed cavity in TE, $I_\nu = B_\nu$ is homogeneous and isotropic, so $dI_\nu/ds = 0$. The RTE then gives $0 = -\alpha_\nu B_\nu + j_\nu$:

$$\boxed{j_\nu = \alpha_\nu B_\nu(T)} \qquad \text{(Kirchhoff's law)}$$

Both j_ν and α_ν depend only on the **state of the matter** (populations and velocity distribution). In LTE that state is the TE state at the local T, so Kirchhoff's law still holds, even though $I_\nu \neq B_\nu$. Hence:

$$\boxed{S_\nu^{\mathrm{LTE}} = \frac{j_\nu}{\alpha_\nu} = B_\nu(T)}$$

MIT 8.901 (§7.3) stresses the other half of this statement: $S_\nu = B_\nu$ does **not** mean the object emits a black-body spectrum. An optically thick lump emits $B_\nu(T)$, but an optically thin lump at the same T emits only $\tau_\nu B_\nu(T)$, which is bright only at the frequencies where α_ν is large (see Lecture 5, §5.4).

Collins (eq. 9.2.28) notes that $S_\nu = B_\nu$ is **sufficient** for LTE but not necessary in general: with electron scattering present, $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$, yet the populations may still be in LTE.

LTE written with Einstein coefficients

With Boltzmann populations, $g_1 n_2 / (g_2 n_1) = e^{-h\nu/kT}$, the formulas of Lecture 4 give

$$\alpha_\nu = \frac{h\nu}{4\pi} n_1 B_{12} \phi(\nu) \left(1 - e^{-h\nu/kT}\right), \qquad \frac{j_\nu}{\alpha_\nu} = \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1} = B_\nu(T). \ \checkmark$$

The factor $\left(1 - e^{-h\nu/kT}\right)$ is the **LTE stimulated-emission correction**; it multiplies every LTE opacity.

What LTE does and does not say

True in LTE	Not implied by LTE
$S_\nu = B_\nu(T_{\rm local})$	$I_\nu = B_\nu$
populations from Saha–Boltzmann at $T_{\rm local}$	$J_\nu = B_\nu$ (it fails near the surface)
opacity and emissivity are functions of (T, ρ) only	isotropic radiation
once $T(\tau)$ is known, the formal solution gives I_ν directly	zero flux

6.4 Boltzmann and Saha equations

Boltzmann (excitation within one ionisation stage):

$$\frac{N_j}{N_i} = \frac{g_j}{g_i}\, e^{-(E_j - E_i)/kT}, \qquad \frac{N_j}{N} = \frac{g_j\, e^{-E_j/kT}}{U(T)}, \quad U = \sum_i g_i e^{-E_i/kT}.$$

For hydrogen, $g_n = 2n^2$ and $E_n = -13.6\ \mathrm{eV}/n^2$.

Saha (ionisation $r \to r+1$):

$$\boxed{\frac{N_{r+1}\, n_e}{N_r} = \frac{2U_{r+1}}{U_r}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_r/kT}}$$

with $(2\pi m_e k/h^2)^{3/2} = 2.415 \times 10^{21}\ \mathrm{m^{-3}\,K^{-3/2}}$. Using $P_e = n_e kT$:

$$\frac{N_{r+1}}{N_r} = \frac{2kT\, U_{r+1}}{P_e\, U_r}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi_r/kT}.$$

The factor 2 is the spin degeneracy of the free electron. The $T^{3/2}$ counts the free-electron phase space, and that large number is why gases ionise at $kT \ll \chi$.

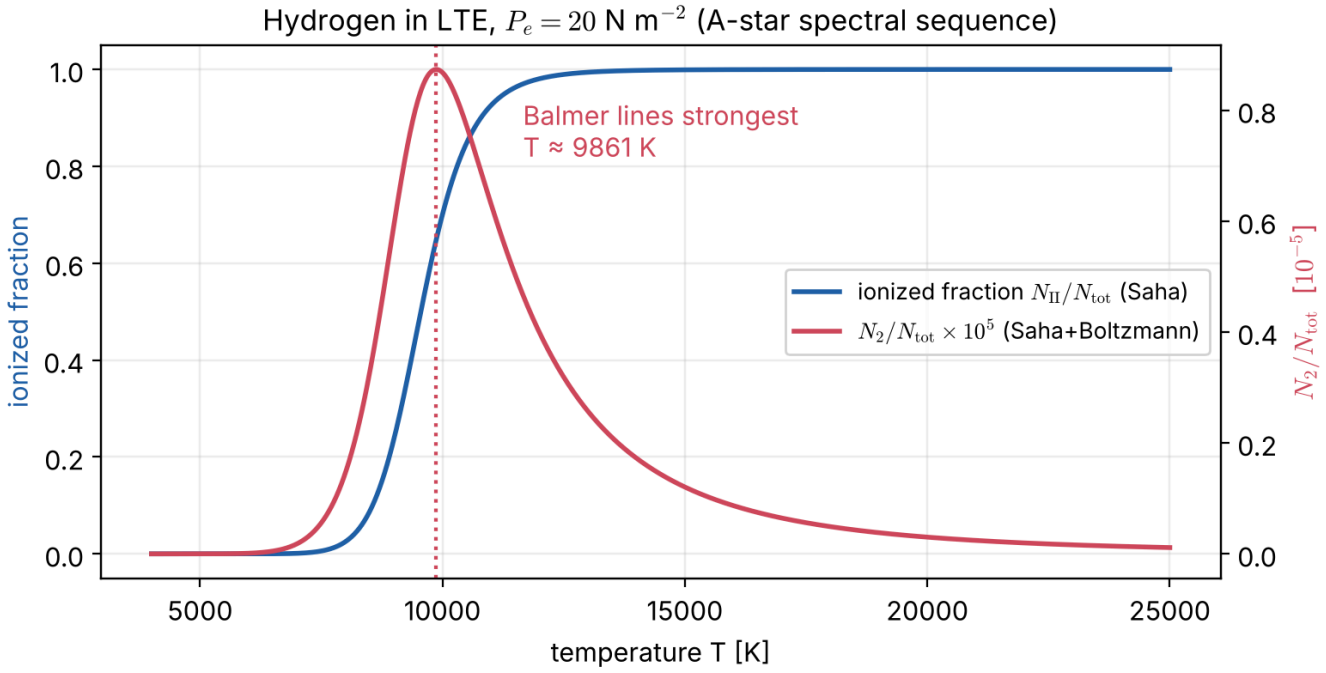


Figure 6.2 — Hydrogen in LTE at $P_e = 20$ N m^{-2}. The $n = 2$ population, which produces the Balmer lines, rises with T (Boltzmann) and then collapses as H ionises (Saha). Its peak at $T \approx 9900$ K explains why Balmer lines are strongest in A0 stars.

6.5 When is LTE valid? The two-level atom

Consider the line $1 \leftrightarrow 2$ with collisional rates C_{12}, C_{21} (per atom per second; $C_{21} = n_e q_{21}$). Detailed balance for collisions gives $C_{12}/C_{21} = (g_2/g_1)e^{-h\nu/kT}$. Statistical equilibrium:

$$n_1 \left(B_{12}\bar{J} + C_{12}\right) = n_2 \left(A_{21} + B_{21}\bar{J} + C_{21}\right).$$

Insert this into $S = (2h\nu^3/c^2)/[g_2 n_1/(g_1 n_2) - 1]$ and use the Einstein relations. After some algebra (Exam Q4):

$$\boxed{S_\nu^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B_\nu(T), \qquad \varepsilon = \frac{\varepsilon'}{1+\varepsilon'}, \quad \varepsilon' = \frac{C_{21}}{A_{21}}\left(1 - e^{-h\nu/kT}\right)}$$

- **Collisions dominate** ($C_{21} \gg A_{21}$, high density): $\varepsilon \to 1$, $S \to B$: **LTE**.
- **Radiation dominates** (low density): $\varepsilon \ll 1$, $S \approx \bar{J}$: scattering-like, **NLTE**.

The **critical density** $n_{\text{crit}} = A_{21}/q_{21}$ separates the two. The line source function has exactly the form found for continuum scattering in Lecture 4, so the methods of Lecture 8 apply.

Collins' five assumptions of "normal" stellar atmospheres (§9.1d)

1. LTE prevails: all material properties follow from the local T and ρ.
2. The atmospheric structure is set by the continuum opacity.
3. The geometry is a plane-parallel slab (thickness $\ll R$).
4. Gravity g is constant through the atmosphere: $dP/dz = -g\rho$.
5. All energy is carried by radiation, with no sources: $F = \sigma T_{\text{eff}}^4$ = constant.

Then **only T_{eff}, g and the composition** specify the atmosphere. This is the basis of Lectures 7–10.

Worked example 6.1 — Why are the Balmer lines weak in the Sun?

At $T = 5772$ K, compute N_2/N_1 for hydrogen. $kT = 0.4974$ eV, so $N_2/N_1 = (8/2)e^{-10.20/0.4974} = 4e^{-20.51} = 5.0 \times 10^{-9}$. Only five atoms in a billion can absorb a Balmer photon. The Balmer lines are still visible because hydrogen is so abundant. At 9900 K: $N_2/N_1 = 4e^{-11.96} = 2.6 \times 10^{-5}$, about 5000 times larger.

Worked example 6.2 — Saha ionisation at 9900 K

Hydrogen at $T = 9900$ K, $P_e = 20$ Pa. Find $N_{\rm II}/N_{\rm I}$. Take $U_{\rm I} = 2$, $U_{\rm II} = 1$.

1. $n_e = P_e/kT = 20/(1.381 \times 10^{-23} \times 9900) = 1.46 \times 10^{20}$ m^{-3}.
2. $(2\pi m_e kT/h^2)^{3/2} = 2.415 \times 10^{21} \times 9900^{3/2} = 2.38 \times 10^{27}$ m^{-3}.
3. $e^{-13.6/0.8531} = e^{-15.94} = 1.19 \times 10^{-7}$.
4. $N_{\rm II}/N_{\rm I} = \dfrac{2 \times 1}{2} \times \dfrac{2.38 \times 10^{27} \times 1.19 \times 10^{-7}}{1.46 \times 10^{20}} = 1.94$.

The ionised fraction is $1.94/2.94 = 0.66$. Then $N_2/N_{\rm tot} = (2.6 \times 10^{-5}/1.000026) \times (1/2.94) \approx 8.7 \times 10^{-6}$, which is the peak in Figure 6.2.

Worked example 6.3 — Is the H I 21-cm line in LTE in the ISM?

$A_{21} = 2.9 \times 10^{-15}$ s^{-1}, and the collisional de-excitation rate coefficient is $q_{21} \approx 10^{-16}$ m^3 s^{-1} (H–H collisions at 100 K). Is the line in LTE at $n_{\rm H} = 10^6$ m^{-3} (1 cm^{-3})?

$n_{\rm crit} = A_{21}/q_{21} \approx 29$ m^{-3} $\ll 10^6$ m^{-3}. Collisions dominate, so the spin temperature is close to the kinetic temperature (the populations are thermalised). By contrast, optical forbidden lines such as [O III] have $A \sim 10^{-2}$ s^{-1} and $n_{\rm crit} \sim 10^{11}$–$10^{12}$ m^{-3}, above nebular densities, so they are strongly NLTE.

Exam questions and answers — Lecture 6

Q1. Define LTE precisely. Explain why LTE can hold even though the radiation field near a stellar surface is very far from a black body.

> **A1.** LTE: at each point the matter (velocity, excitation and ionisation distributions) is in TE at the local temperature $T(\mathbf{r})$. It can hold because particle–particle collisions, which set the material distributions, are frequent (short particle mean free path). Photons have long mean free paths and escape, so $I_\nu \neq B_\nu$, but as long as collisional rates dominate the radiative rates in setting the populations, the matter does not "notice" the anomalous radiation field.

Q2. Derive Kirchhoff's law and show that it implies $S_\nu = B_\nu$ in LTE.

> **A2.** In a TE cavity, $I_\nu = B_\nu$ is uniform and isotropic, so $dI_\nu/ds = 0 = j_\nu - \alpha_\nu B_\nu$, giving $j_\nu = \alpha_\nu B_\nu(T)$. Because j_ν and α_ν are properties of the matter only, the relation holds whenever the matter is in its TE state at the local T, i.e. in LTE. Then $S_\nu = j_\nu/\alpha_\nu = B_\nu(T)$.

Q3. Explain, with Saha and Boltzmann, why Balmer lines peak in strength for A0 stars ($T \approx 10^4$ K) and are weaker in both cooler and hotter stars.

A3. The Balmer opacity $\propto N_2$. In cool stars, $N_2/N_1 \propto e^{-10.2\,\mathrm{eV}/kT}$ is tiny (e.g. 5×10^{-9} at 5800 K). Above about 10^4 K, Saha ionisation removes neutral H: $N_\mathrm{I}/N_\mathrm{tot} \to 0$. The product N_2/N_tot therefore peaks near 9900 K (for $P_e = 20$ Pa).

Q4. For a two-level atom with collisions, derive $S = (1-\varepsilon)\bar{J} + \varepsilon B$. When does the line form in LTE?

A4. From statistical equilibrium, $\frac{n_1}{n_2} = \frac{A_{21}+B_{21}\bar{J}+C_{21}}{B_{12}\bar{J}+C_{12}}$. Then $\frac{g_2 n_1}{g_1 n_2} - 1 = \frac{A_{21}+C_{21}(1-e^{-h\nu/kT})}{B_{21}\bar{J}+C_{21}e^{-h\nu/kT}}$ (using $g_2 B_{21} = g_1 B_{12}$ and $g_2 C_{21} e^{-h\nu/kT} = g_1 C_{12}$). So $S = \frac{2h\nu^3}{c^2}\,\frac{B_{21}\bar{J}+C_{21}e^{-h\nu/kT}}{A_{21}+C_{21}(1-e^{-h\nu/kT})}$. With $A_{21} = (2h\nu^3/c^2)B_{21}$ and $\varepsilon' = C_{21}(1-e^{-h\nu/kT})/A_{21}$: $S = \frac{\bar{J}+\varepsilon' B_\nu}{1+\varepsilon'}$, because $\frac{2h\nu^3}{c^2}\,\frac{e^{-h\nu/kT}}{1-e^{-h\nu/kT}} = B_\nu$. Writing $\varepsilon = \varepsilon'/(1+\varepsilon')$ gives the result. LTE holds when $C_{21} \gg A_{21}$, i.e. $n_e \gg n_\mathrm{crit}$, but **also** only at depths $\gtrsim 1/\sqrt{\varepsilon}$ (Lecture 8).

Lecture 7 — Plane-Parallel Atmospheres: Schwarzschild–Milne Equations and Eddington–Barbier

> **Sources:** Collins §9.2 (plane-parallel RTE, eq. 9.2.24–9.2.26, Fig. 9.3), §9.3 (moments J, H, K); Chandrasekhar Ch. I §§9–11 (the formal solution and the Λ, Φ, X operators); Hu Set 1 (formal solution); MIT 8.901 §7.2.

Learning goals

1. Write the RTE in plane-parallel geometry with τ measured **inward**.
2. Derive the formal solutions for outgoing and incoming intensity and the Schwarzschild–Milne integrals for J, H and K.
3. Prove the Eddington–Barbier relations and use them to invert limb darkening into a temperature structure.

7.1 Geometry and sign conventions

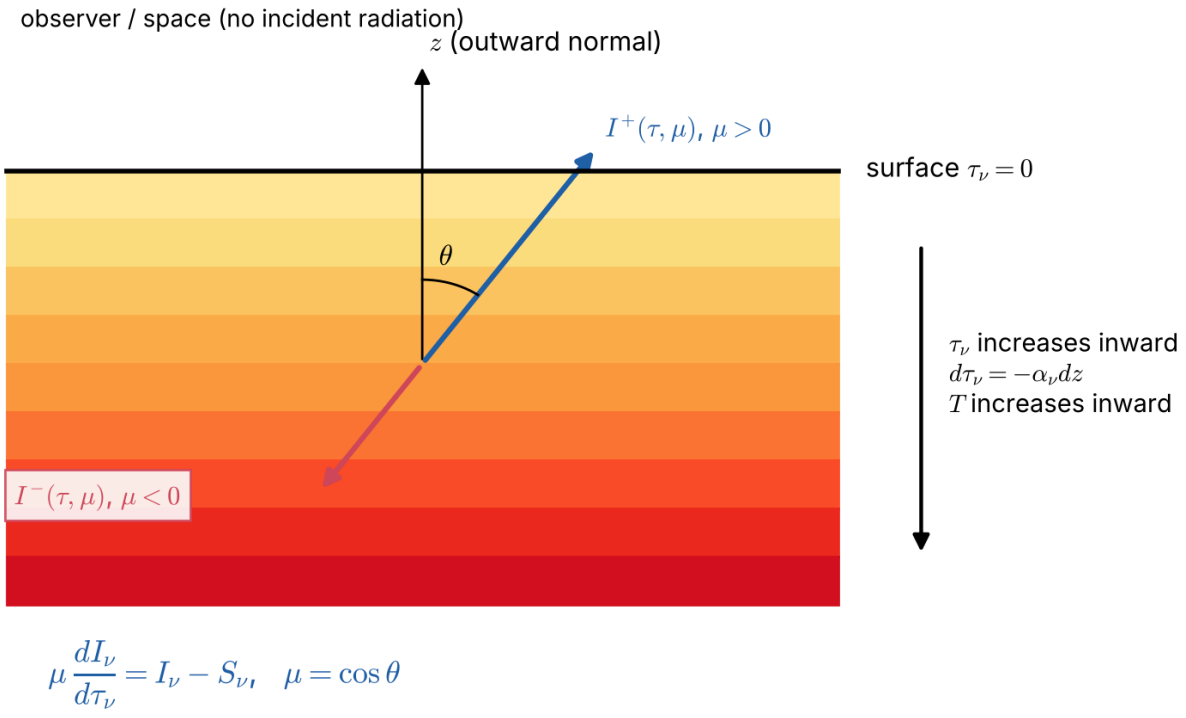


*Figure 7.1 — A plane-parallel atmosphere. z points outward, θ is measured from the outward normal, and $\mu=\cos\theta$. Outgoing rays have $\mu>0$, incoming rays $\mu<0$. The optical depth increases **inward**: $d\tau_\nu=-\alpha_\nu dz$.*

Along a ray, $ds=dz/\mu$. Then $dI_\nu/ds=\mu\,dI_\nu/dz=-\mu\,\alpha_\nu\,dI_\nu/d\tau_\nu$, and the RTE becomes (Collins eq. 9.2.26)

$$\boxed{\mu \frac{dI_\nu(\tau_\nu,\mu)}{d\tau_\nu} = I_\nu(\tau_\nu,\mu) - S_\nu(\tau_\nu)}$$

Note the sign change compared with Lecture 5, because τ is now measured against the outgoing direction.

Boundary conditions for a semi-infinite stellar atmosphere:

- no incident radiation at the top: $I_\nu(0,\mu<0)=0$;
- the intensity does not grow faster than e^{τ} at depth: $\lim_{\tau\to\infty} I_\nu e^{-\tau/\mu} = 0$.

7.2 Formal solution in a semi-infinite atmosphere

Multiply by $e^{-\tau/\mu}/\mu$: $\dfrac{d}{d\tau}\left(I e^{-\tau/\mu}\right) = -\dfrac{S}{\mu} e^{-\tau/\mu}$.

Outgoing ($0<\mu\le 1$), integrating from τ to ∞:

$$I_\nu^{+}(\tau,\mu) = \int_\tau^\infty S_\nu(t)\, e^{-(t-\tau)/\mu}\, \frac{dt}{\mu}.$$

Incoming ($\mu<0$; write $\mu=-|\mu|$), integrating from 0 to τ with $I(0)=0$:

$$I_\nu^{-}(\tau,\mu) = \int_0^\tau S_\nu(t)\, e^{-(\tau-t)/|\mu|}\, \frac{dt}{|\mu|}.$$

Emergent intensity (what we observe at disk position μ):

$$\boxed{I_\nu(0,\mu) = \int_0^\infty S_\nu(t)\, e^{-t/\mu}\, \frac{dt}{\mu}}$$

This is a **Laplace transform** of $S_\nu(\tau)$. Measuring $I_\nu(0,\mu)$ across the disk of the Sun (μ from 1 at centre to 0 at the limb) therefore probes $S_\nu(\tau)$ at different depths.

7.3 Exponential integrals and the Schwarzschild–Milne equations

$$E_n(x) = \int_1^\infty \frac{e^{-xt}}{t^n}\, dt = \int_0^1 e^{-x/\mu}\mu^{n-2}\, d\mu, \qquad \frac{dE_n}{dx} = -E_{n-1}, \quad E_n(0) = \frac{1}{n-1}\ (n>1).$$

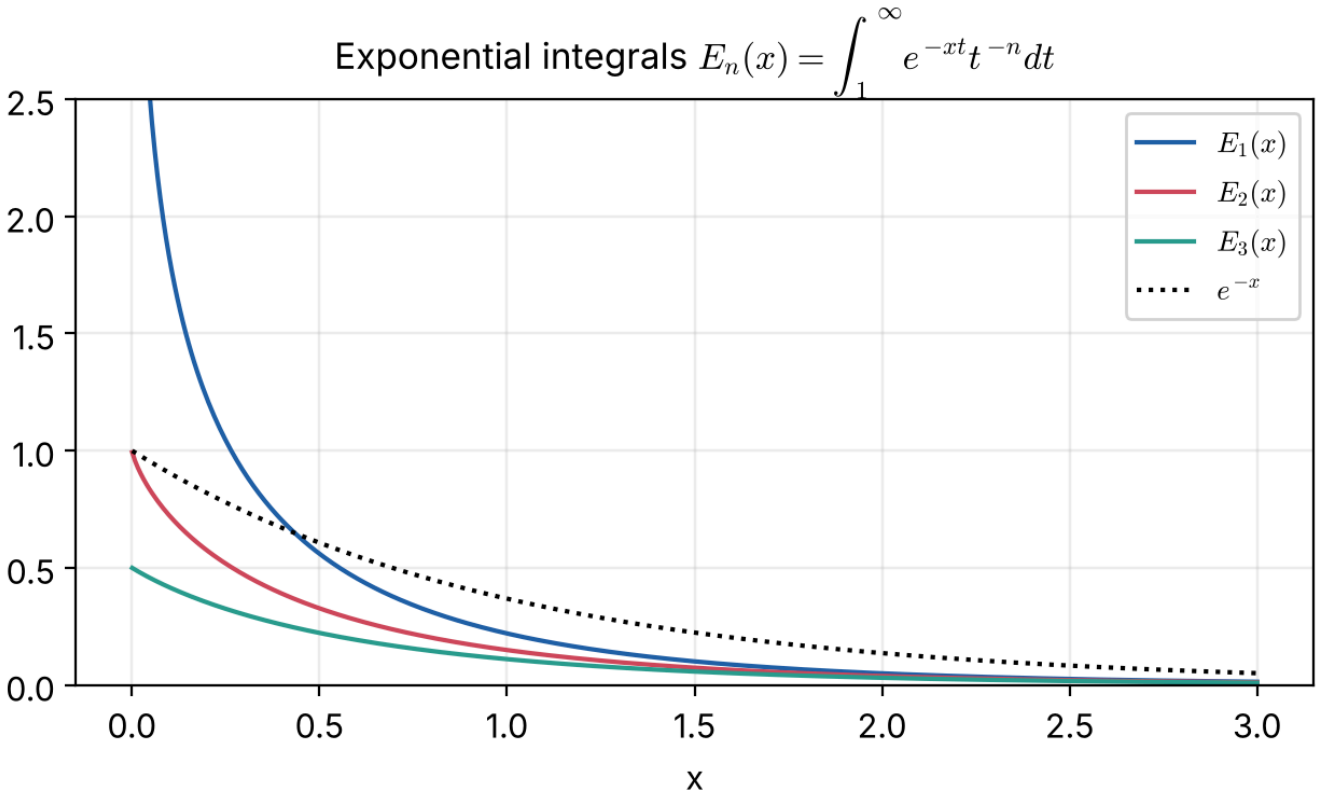


Figure 7.2 — E_1 diverges logarithmically at $x = 0$ (but is integrable); all E_n decay roughly like e^{-x}/x.

Insert the formal solution into the moment definitions and swap the order of integration:

$$J_\nu(\tau) = \tfrac{1}{2}\int_0^\infty S_\nu(t)\, E_1|t-\tau|\, dt \;\equiv\; \Lambda_\tau[S_\nu]$$

$$F_\nu(\tau) = 2\pi\left[\int_\tau^\infty S_\nu(t) E_2(t-\tau)\, dt - \int_0^\tau S_\nu(t) E_2(\tau - t)\, dt\right] \;\equiv\; \Phi_\tau[S_\nu]$$

$$K_\nu(\tau) = \tfrac{1}{2}\int_0^\infty S_\nu(t)\, E_3|t-\tau|\, dt \;\equiv\; X_\tau[S_\nu]$$

These are the **Schwarzschild–Milne equations**, and Λ is the **lambda operator** of Chandrasekhar.

Example: S constant. $J(0) = \tfrac{1}{2} S \int_0^\infty E_1(t) dt = \tfrac{1}{2} S\, E_2(0) = S/2$. At the surface only half the sky is filled, so $J = S/2$. Deep inside, $J \to S$.

7.4 The Eddington–Barbier relations

Expand the source function near the surface as $S_\nu(\tau) = a + b\tau$. Then

$$I_\nu(0,\mu) = \int_0^\infty (a + bt) e^{-t/\mu} \frac{dt}{\mu} = a + b\mu,$$

using $\int_0^\infty e^{-t/\mu} dt/\mu = 1$ and $\int_0^\infty t e^{-t/\mu} dt/\mu = \mu$. So

$$\boxed{I_\nu(0,\mu) = S_\nu(\tau_\nu = \mu)}$$

exactly for a linear S, and approximately otherwise. For the surface flux:

$$F_\nu(0) = 2\pi \int_0^1 (a + b\mu)\mu \, d\mu = \pi \left(a + \tfrac{2}{3} b\right) \;\Rightarrow\; \boxed{F_\nu(0) = \pi S_\nu(\tau_\nu = 2/3)}$$

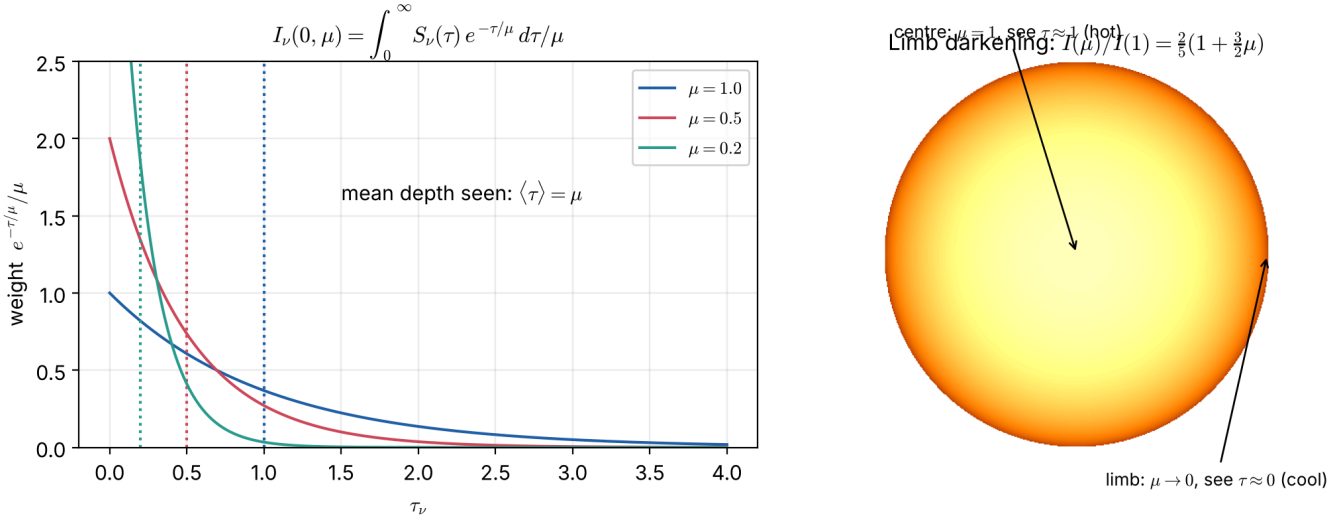


Figure 7.3 — Left: the weighting kernel $e^{-\tau/\mu}/\mu$ peaks at the surface but has mean depth $\langle\tau\rangle = \mu$. Right: limb darkening. At disk centre we see down to $\tau \approx 1$ (hot); at the limb, rays graze the atmosphere and we see $\tau \approx 0$ (cool).

Physical meaning:

- "We see down to $\tau \approx 1$ along the line of sight," i.e. to vertical depth $\tau \approx \mu$.
- Toward the limb we see higher, cooler layers. If S increases inward ($b > 0$), the limb is **darker**.
- The observed flux of a star reflects the conditions at $\tau \approx 2/3$, which is why the "photosphere" is defined there and why $T(\tau = 2/3) \approx T_{\text{eff}}$ (Lecture 9).

7.5 From limb darkening to temperature: an LTE inversion

In LTE, $S_\nu = B_\nu(T)$. Measuring $I_\nu(0,\mu)$ gives $S_\nu(\tau_\nu = \mu)$ by Eddington–Barbier, and inverting the Planck function then gives $T(\tau_\nu)$. This is how the first empirical models of the solar photosphere were built.

Worked example 7.1 — The temperature of the solar photosphere from limb darkening

At $\lambda = 500$ nm the solar limb darkening is approximately $I(\mu)/I(1) = 0.36 + 0.64\mu$, and the disk-centre intensity is $I(1) \approx 4.0 \times 10^{13}$ W m^{-3} sr^{-1}. Assume LTE. Find T at $\tau_{500} = 0,\, 0.5$ and 1.

Step 1 (Eddington–Barbier): $S(\tau) = I(1)\,(0.36 + 0.64\tau)$, so $S(0) = 0.36 I(1) = 1.44 \times 10^{13}$, $S(0.5) = 0.68 I(1) = 2.72 \times 10^{13}$, $S(1) = 4.0 \times 10^{13}$.

Step 2 (LTE): $S = B_\lambda(T)$, so $T = \dfrac{hc/\lambda k}{\ln(1 + 2hc^2/\lambda^5 S)}$ with $hc/\lambda k = 28\,776$ K and $2hc^2/\lambda^5 = 3.811 \times 10^{15}$:

τ_{500}	$S/I(1)$	T [K]
0	0.36	5155
0.5	0.68	5814
1	1.00	6300

Conclusion: the temperature **increases inward**, by roughly 1100 K over one optical depth. This is direct observational proof that the Sun is not isothermal, and it explains why the solar spectrum has absorption lines.

Worked example 7.2 — Emergent intensity for a quadratic source function

$S(\tau) = a + b\tau + c\tau^2$. *Find* $I(0,\mu)$ *and the error of Eddington–Barbier.*

$\int_0^\infty t^2 e^{-t/\mu} dt/\mu = 2\mu^2$, so $I(0,\mu) = a + b\mu + 2c\mu^2$, whereas $S(\mu) = a + b\mu + c\mu^2$. The error is $c\mu^2$, i.e. Eddington–Barbier is exact only for linear S.

Worked example 7.3 — Mean intensity at the surface for linear S

$S = a + b\tau$. *Compute* $J(0)$ *and* $H(0)$*, and show that* $J(0) \neq S(0)$.

$I^+(0,\mu) = a + b\mu$ and $I^-(0) = 0$, so $J(0) = \frac{1}{2}\int_0^1 (a + b\mu) d\mu = \frac{a}{2} + \frac{b}{4}$ and $H(0) = \frac{1}{2}\int_0^1 (a + b\mu)\mu \, d\mu = \frac{a}{4} + \frac{b}{6}$. With $b = 0$: $J(0) = a/2 = S(0)/2$. **The radiation field at the surface is not in equilibrium with the local source function**, a signature of photon escape.

Exam questions and answers — Lecture 7

Q1. Derive the formal solution for the emergent intensity of a semi-infinite atmosphere.

> **A1.** $\frac{d}{d\tau}(Ie^{-\tau/\mu}) = -\frac{S}{\mu}e^{-\tau/\mu}$. Integrate from 0 to ∞ with $Ie^{-\tau/\mu} \to 0$: $-I(0,\mu) = -\int_0^\infty S e^{-t/\mu} dt/\mu$, so $I(0,\mu) = \int_0^\infty S(t) e^{-t/\mu} dt/\mu$.

Q2. Prove the Eddington–Barbier relations $I(0,\mu) = S(\tau = \mu)$ and $F(0) = \pi S(\tau = 2/3)$ for $S = a + b\tau$.

> **A2.** See §7.4: $I(0,\mu) = a + b\mu = S(\mu)$; $F = 2\pi\int_0^1 (a + b\mu)\mu d\mu = 2\pi(a/2 + b/3) = \pi(a + 2b/3) = \pi S(2/3)$.

Q3. Explain limb darkening physically. Under what condition would we see limb **brightening**?

> **A3.** At disk centre the line of sight is perpendicular to the layers, so $\tau = 1$ is reached at vertical depth $\tau_{\text{vert}} = 1$. Near the limb the ray is oblique, and $\tau = 1$ along it occurs at $\tau_{\text{vert}} = \mu$, in higher layers. If S (in LTE, T) decreases outward, the limb is darker. If S increases outward, as in the chromosphere at UV, EUV and radio wavelengths, the limb is brighter.

Q4. The solar limb darkening at some wavelength is $I(\mu)/I(1) = 0.3 + 0.7\mu$. In LTE and using Eddington–Barbier, at what optical depth does S equal half its value at $\tau = 1$?

> **A4.** $S(\tau)/S(1) = 0.3 + 0.7\tau = 0.5 \Rightarrow \tau = 0.29$.

Lecture 8 — Moment Equations, the Diffusion and Eddington Approximations, and Scattering Atmospheres

> **Sources:** Collins §9.3–9.4 (moments; radiative equilibrium eq. 9.4.3–9.4.5; diffusion approximation eq. 9.4.7–9.4.9; Eddington approximation eq. 9.4.12–9.4.13); Hu Set 1 (scattering, multiple processes, single-scattering albedo); Chandrasekhar Ch. I §§11–14 and Ch. III (isotropic scattering).

Learning goals

1. Derive the zeroth and first moment equations and the condition of radiative equilibrium.
2. Close the moment hierarchy with the Eddington approximation and its surface boundary condition.
3. Derive the diffusion approximation and the Rosseland mean opacity.
4. Solve an isothermal atmosphere with scattering, obtain the $\sqrt{\varepsilon}$ law and the thermalisation depth, and understand why Λ-iteration fails.

8.1 Moments of the RTE

Take $\mu\, dI_\nu/d\tau_\nu = I_\nu - S_\nu$ (S_ν isotropic) and apply $\frac{1}{2}\int_{-1}^{1} d\mu$ and $\frac{1}{2}\int_{-1}^{1} \mu\, d\mu$:

$$\boxed{\frac{dH_\nu}{d\tau_\nu} = J_\nu - S_\nu} \qquad \boxed{\frac{dK_\nu}{d\tau_\nu} = H_\nu}$$

Each equation brings in a higher moment ($J \to H \to K \to \ldots$): the **closure problem**, as in the hydrodynamic moments of the Boltzmann equation (Collins §9.4).

Radiative equilibrium

If radiation carries all the energy and there are no sources, the total flux is constant: $dF/dz = 0$ (Collins eq. 9.4.5). Integrating the zeroth moment over frequency:

$$\frac{dH}{d\tau} \;\propto\; \int_0^\infty \alpha_\nu \left(J_\nu - S_\nu\right) d\nu = 0.$$

In LTE ($S_\nu = B_\nu$ plus coherent scattering, which cancels out because scattering conserves energy):

$$\boxed{\int_0^\infty \kappa_\nu\, J_\nu\, d\nu = \int_0^\infty \kappa_\nu\, B_\nu(T)\, d\nu} \qquad \text{(energy absorbed = energy emitted)}$$

This condition **determines** $T(\tau)$. It is the extra equation that turns the RTE into a model atmosphere.

8.2 The diffusion approximation (deep interior)

At large depth, $S_\nu = B_\nu$ varies slowly on the scale of a mean free path. Expanding $B_\nu(t)$ in a Taylor series about τ in the formal solution gives

$$I_\nu(\tau,\mu) = B_\nu + \mu\frac{dB_\nu}{d\tau_\nu} + \mu^2\frac{d^2B_\nu}{d\tau_\nu^2} + \dots$$

The radiation is almost isotropic with a small linear anisotropy. Then (cf. Lecture 2, Worked example 2.3)

$$J_\nu \approx B_\nu, \qquad H_\nu = \frac{1}{3}\frac{dB_\nu}{d\tau_\nu}, \qquad K_\nu = \frac{B_\nu}{3} \approx \frac{J_\nu}{3}.$$

The flux is therefore

$$F_\nu = 4\pi H_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} = -\frac{4\pi}{3}\frac{1}{\alpha_\nu}\frac{\partial B_\nu}{\partial T}\frac{dT}{dz}.$$

Radiation **diffuses** down the temperature gradient, like heat conduction.

Rosseland mean opacity

Integrate over frequency:

$$F = -\frac{4\pi}{3}\frac{dT}{dz}\int_0^\infty \frac{1}{\alpha_\nu}\frac{\partial B_\nu}{\partial T}d\nu \equiv -\frac{4\pi}{3}\frac{1}{\alpha_R}\frac{dB}{dT}\frac{dT}{dz},$$

which defines the **Rosseland mean**

$$\boxed{\frac{1}{\alpha_R} = \frac{\displaystyle\int_0^\infty \frac{1}{\alpha_\nu}\frac{\partial B_\nu}{\partial T}d\nu}{\displaystyle\int_0^\infty \frac{\partial B_\nu}{\partial T}d\nu}} \quad\Rightarrow\quad \boxed{F = -\frac{16\sigma T^3}{3\kappa_R\rho}\frac{dT}{dz}}$$

using $dB/dT = 4\sigma T^3/\pi$. It is a **harmonic** mean: energy flows through the frequency "windows" where the opacity is low (Hu: "energy escapes in the channel with the lowest opacity"). This is the radiative-transfer equation of **stellar interiors**.

8.3 The Eddington approximation (whole atmosphere)

Eddington assumed $K_\nu = J_\nu/3$ **everywhere**, not only at depth (Collins eq. 9.4.8). At the surface we also need a boundary condition. Assume the outgoing intensity is isotropic, $I(0,\mu>0) = I_0$, with $I(0,\mu<0) = 0$. Then $J(0) = I_0/2$ and $H(0) = I_0/4$ (Lecture 2, Q3), so

$$\boxed{J(0) = 2H(0)} \qquad \text{(Eddington boundary condition; Collins eq. 9.4.13)}$$

The two errors partly cancel, and the approximation is typically good to about 10 % (Collins), and to a few per cent for the temperature of a gray atmosphere (Lecture 9: $T(0)$ is $0.841\,T_{\rm eff}$ versus the exact $0.811\,T_{\rm eff}$).

Combined, the moment equations become a **second-order ODE**:

$$\frac{1}{3}\frac{d^2 J_\nu}{d\tau_\nu^2} = J_\nu - S_\nu .$$

8.4 Atmosphere with scattering: thermalisation

Take the source function of Lecture 4,

$$S = \varepsilon B + (1-\varepsilon) J, \qquad J - S = \varepsilon (J - B),$$

in an **isothermal** atmosphere (B = constant), with ε constant.

Eddington solution

$\frac{1}{3} J'' = \varepsilon (J - B)$, so $J = B + C e^{-\sqrt{3\varepsilon}\,\tau}$ (the growing solution is excluded). The boundary condition $J(0) = 2H(0) = \frac{2}{3} J'(0)$ gives $B + C = -\frac{2}{3}\sqrt{3\varepsilon}\, C$, hence

$$J(\tau) = B\left[1 - \frac{e^{-\sqrt{3\varepsilon}\,\tau}}{1 + 2\sqrt{\varepsilon/3}}\right], \qquad S(0) = B\left[\varepsilon + (1-\varepsilon)\frac{2\sqrt{\varepsilon/3}}{1 + 2\sqrt{\varepsilon/3}}\right] \approx \frac{2}{\sqrt{3}}\sqrt{\varepsilon}\, B = 1.15\sqrt{\varepsilon}\, I$$

Exact result

The exact solution of the integral equation $S = \varepsilon B + (1-\varepsilon)\Lambda[S]$ (Chandrasekhar's H-function theory) gives the famous

$$\boxed{S(0) = \sqrt{\varepsilon}\, B}$$

This is one of the most important results in radiative transfer.

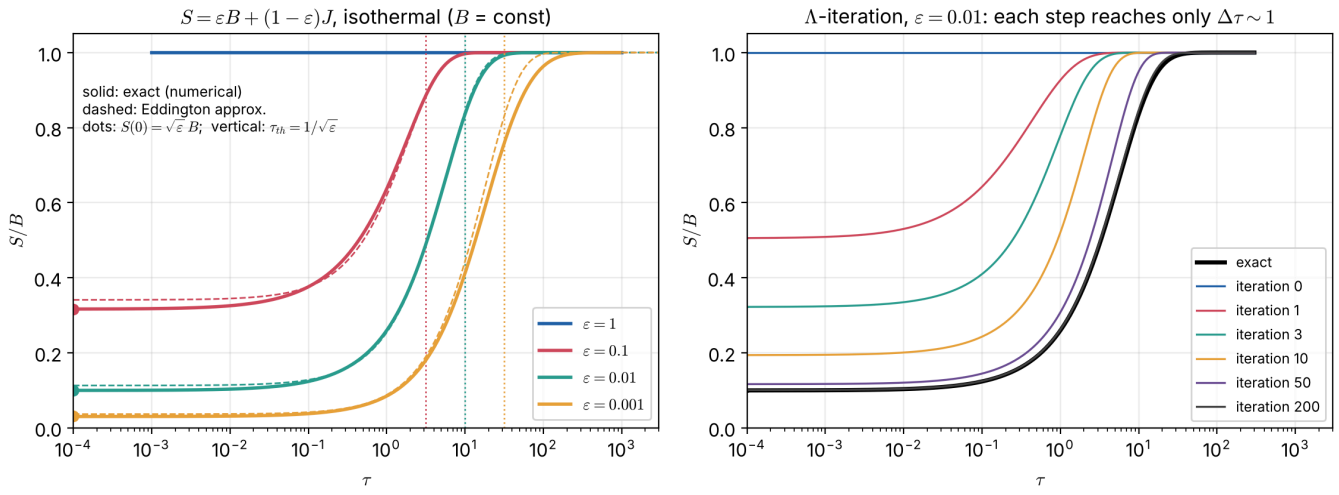


Figure 8.1 — Left: exact numerical solutions (solid; computed with `scripts/lte_solvers.py`) and the Eddington approximation (dashed) for an isothermal atmosphere with $S = \varepsilon B + (1-\varepsilon)J$. Even though B is constant, S drops far below B near the surface, and $S \to B$ only below the thermalisation depth $\tau_{\rm th} \sim 1/\sqrt{\varepsilon}$ (dotted lines). Right: Λ-iteration for $\varepsilon = 0.01$. Each iteration propagates information only about one optical depth; after 200 iterations it is still not converged.

Interpretation.

- A photon is destroyed with probability ε per interaction, so it survives $\sim 1/\varepsilon$ scatterings and random-walks a distance $\Delta\tau \sim \sqrt{1/\varepsilon}$. That is the **thermalisation length**.
- Within $\tau_{\rm th}$ of the surface, photons "know" about the boundary and leak out. J drops below B, so $S < B$: **NLTE behaviour even when the gas is isothermal and in LTE populations.**
- For strong resonance lines $\varepsilon \sim 10^{-4}$–10^{-8}, so $\tau_{\rm th} \sim 10^{2}$–$10^{4}$. Thermalisation happens very deep, and line cores are much darker than LTE predicts.

Why Λ-iteration fails

The obvious scheme $S^{(n+1)} = \varepsilon B + (1-\varepsilon)\Lambda[S^{(n)}]$ converges at rate $(1-\varepsilon)$ per step. Each step only couples points within $\Delta\tau \sim 1$. For $\varepsilon \ll 1$ it can appear converged (tiny changes) while still being badly wrong (Figure 8.1, right). Modern codes use **accelerated Λ-iteration (ALI)**, or solve the linear system directly, as our script does: $(\mathbf{1} - (1-\varepsilon)\mathbf{\Lambda})\mathbf{S} = \varepsilon\mathbf{B}$.

Numerical solution: how the script works

1. Choose a grid $\tau_0 = 0 < \tau_1 < \cdots < \tau_N$ (logarithmic).
2. Represent $S(\tau)$ as piecewise linear between grid points.
3. Integrate $J_i = \tfrac{1}{2}\int S(t) E_1|t-\tau_i|\,dt$ analytically on each segment, using $\int E_1 = -E_2$ and $\int x E_1\,dx = -xE_2 - E_3$. This gives a matrix Λ_{ik} with $J_i = \sum_k \Lambda_{ik} S_k$.
4. Solve the linear system for S_k.

The results reproduce $S(0)/B = 0.316,\ 0.099$ for $\varepsilon = 0.1,\ 0.01$, against exact values of 0.316 and 0.100.

Worked example 8.1 — Rosseland mean of a two-band opacity

Suppose the weight $\partial B_\nu/\partial T$ is split equally between two bands with opacities $\kappa_1 = 1$ and $\kappa_2 = 100$ (arbitrary units). Compare the Rosseland mean with the straight (Planck-like) mean.

$$\frac{1}{\kappa_R} = \frac{1}{2}\left(\frac{1}{1} + \frac{1}{100}\right) = 0.505 \;\Rightarrow\; \kappa_R = 1.98; \qquad \kappa_{\rm Planck} = \tfrac{1}{2}(1+100) = 50.5.$$

The flux flows through the transparent band. Adding many strong lines (line blanketing) raises κ_R only modestly, unless they fill the windows.

Worked example 8.2 — Thermalisation of a resonance line

A resonance line in a stellar chromosphere has $\varepsilon = 10^{-4}$. Estimate (a) $S(0)/B$, (b) the thermalisation depth, (c) the number of scatterings before destruction.

a. $S(0) = \sqrt{\varepsilon}\,B = 0.01\,B$. (b) $\tau_{\rm th} \approx 1/\sqrt{\varepsilon} = 100$ (Eddington: $1/\sqrt{3\varepsilon} = 58$). (c) $N \approx 1/\varepsilon = 10^{4}$. **Consequence:** an LTE calculation ($S = B$) would predict a line core 100 times brighter than the NLTE one. This is why strong lines like Ca II H & K and Mg II h & k need NLTE treatment.

Worked example 8.3 — Diffusion flux and the solar interior

At $r \approx 0.5 R_\odot$ in the Sun, $T \approx 3.4\times10^{6}$ K, $\rho \approx 1.3\times10^{3}$ kg m^{-3} and $\kappa_R \approx 1$ m^2 kg^{-1}, and the luminosity passing through is $L_\odot$. Find dT/dr.

$$F = L_\odot/(4\pi r^2) = 3.828\times10^{26}/(4\pi(3.48\times10^{8})^2) = 2.5\times10^{8}\ \mathrm{W\,m^{-2}}.$$

$$\left|\frac{dT}{dr}\right| = \frac{3\kappa_R\rho F}{16\sigma T^3} = \frac{3\times1\times1.3\times10^{3}\times2.5\times10^{8}}{16\times5.67\times10^{-8}\times(3.4\times10^{6})^3} = 2.7\times10^{-2}\ \mathrm{K\,m^{-1}}.$$

That is about 27 K per km, or roughly $T/R_\odot \times 5$: the right order of magnitude for the solar interior.

Exam questions and answers — Lecture 8

Q1. State radiative equilibrium in integral form. Why do scattering terms drop out?

> **A1.** $\int \alpha_\nu(J_\nu - S_\nu)d\nu = 0$, equivalently $dF/dz = 0$. For coherent scattering the scattering part of $\alpha_\nu(J_\nu - S_\nu)$ is $\alpha_\nu^{\mathrm{sc}}(J_\nu - J_\nu) = 0$: scattering redirects photons but neither creates nor destroys energy (Collins §9.4). The condition becomes $\int \kappa_\nu^a(J_\nu - B_\nu)d\nu = 0$.

Q2. Derive the diffusion approximation and the Rosseland mean opacity. Why is it a harmonic mean?

> **A2.** See §8.2. Deep inside, $I \approx B + \mu\, dB/d\tau$, so $F_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} = -\frac{4\pi}{3\alpha_\nu}\frac{\partial B_\nu}{\partial T}\frac{dT}{dz}$. Summing the fluxes over frequencies adds the $1/\alpha_\nu$ ("conductances", like resistors in parallel), hence the harmonic mean weighted by $\partial B_\nu/\partial T$.

Q3. For an isothermal atmosphere with $S = \varepsilon B + (1-\varepsilon)J$, solve in the Eddington approximation for $J(\tau)$. Explain why $S \neq B$ near the surface although the atmosphere is isothermal.

> **A3.** $\frac{1}{3}J'' = \varepsilon(J-B) \Rightarrow J = B + Ce^{-\sqrt{3\varepsilon}\tau}$. With $J(0) = \frac{2}{3}J'(0)$: $C = -B/(1+2\sqrt{\varepsilon/3})$. Near the surface photons escape, so $J < B$; scattering makes S depend on J, so S is also less than B. Only below $\tau \sim 1/\sqrt{\varepsilon}$ do photons get destroyed (thermalised) before they can escape, and there $S \to B$.

Q4. Why does Λ-iteration converge so slowly when $\varepsilon \ll 1$? Name a better approach.

> **A4.** The Λ operator couples points only over $\Delta\tau \sim 1$, so each iteration spreads information about the boundary by about one optical depth. The thermalisation depth is $\sim 1/\sqrt{\varepsilon}$, and the error decays as $(1-\varepsilon)^n$, so $\gtrsim 1/\varepsilon$ iterations are needed. Remedies: solve the linear system directly (Feautrier method, or matrix inversion as in our script), or use accelerated Λ-iteration (ALI) with an approximate local operator.

Lecture 9 — The Gray LTE Atmosphere: Complete Solution

This is the central lecture of the course. We solve, step by step and then exactly, the simplest self-consistent model of a stellar atmosphere in LTE. **Sources:** Collins §9.1d (assumptions), §9.4 (radiative equilibrium; Eddington approximation, eq. 9.4.12–9.4.13); Chandrasekhar Ch. III §25 (Milne problem by discrete ordinates: $J(\tau)=\frac{3}{4}F[\tau+q(\tau)]$, eq. 45; H-function law of darkening, eq. 61; Hopf–Bronstein relation $J(0)=\frac{\sqrt{3}}{4}F$, eq. 64) and Ch. V (exact tables); Hu Set 1; MIT 8.901.

Learning goals

1. List the assumptions of the gray LTE atmosphere and explain each.
2. Derive $T^4(\tau)=\frac{3}{4}T_{\rm eff}^4\left(\tau+\frac{2}{3}\right)$ in the Eddington approximation.
3. Know the exact solution (Hopf function $q(\tau)$, Chandrasekhar's H-function) and its numerical values.
4. Compute the emergent intensity, limb darkening, spectrum and pressure structure of a gray LTE model.

9.1 The problem

Assumptions

1. **Plane-parallel, static** atmosphere; τ measured inward.
2. **LTE:** $S_\nu = B_\nu(T)$ (pure absorption; scattering can be added and changes nothing for the gray case, since it cancels in radiative equilibrium).
3. **Gray:** $\kappa_\nu=\kappa$ is independent of frequency, so $\tau_\nu=\tau$ for all ν. (In practice we use the Rosseland mean: Lecture 8.)
4. **Radiative equilibrium:** constant flux $F=\sigma T_{\rm eff}^4$, with no convection and no energy sources.

Frequency-integrated equations. Integrating $\mu\,dI_\nu/d\tau = I_\nu - B_\nu$ over ν (allowed because τ does not depend on ν):

$$\mu\frac{dI}{d\tau} = I - B, \qquad B=\frac{\sigma T^4}{\pi}.$$

9.2 Step-by-step solution in the Eddington approximation

Step 1 — Radiative equilibrium gives $J=B$. Radiative equilibrium, $\int\kappa_\nu(J_\nu-B_\nu)d\nu=0$, becomes for a gray atmosphere $\kappa(J-B)=0$:

$$\boxed{J(\tau)=B(\tau)=S(\tau)}$$

In a gray LTE atmosphere the source function equals the mean intensity. This is mathematically identical to a **pure-scattering** atmosphere (the Milne problem), which is why Chandrasekhar treats both together.

Step 2 — The flux is constant. The zeroth moment gives $dH/d\tau = J - B = 0$, so

$$H = \text{const} = \frac{F}{4\pi} = \frac{\sigma T_{\text{eff}}^4}{4\pi}.$$

Step 3 — K is linear in τ (exactly). The first moment, $dK/d\tau = H$, gives

$$K(\tau) = H\tau + \text{const}.$$

No approximation has been made so far.

Step 4 — Eddington closure. With $K = J/3$:

$$J(\tau) = 3H\,(\tau + C).$$

Step 5 — Surface boundary condition. With $J(0) = 2H(0)$ (Lecture 8): $3HC = 2H$, so $C = 2/3$:

$$J(\tau) = 3H\left(\tau + \frac{2}{3}\right) = \frac{3\sigma T_{\text{eff}}^4}{4\pi}\left(\tau + \frac{2}{3}\right).$$

Step 6 — Temperature. Since $J = B = \sigma T^4/\pi$:

$$\boxed{T^4(\tau) = \frac{3}{4}\,T_{\text{eff}}^4\left(\tau + \frac{2}{3}\right)}$$

Step 7 — Consequences.

Quantity	Value	Meaning
$T(\tau = 2/3)$	T_{eff}	the "photosphere": the effective temperature is the actual temperature at $\tau = 2/3$ (Eddington–Barbier)
$T(0)$	$2^{-1/4}T_{\text{eff}} = 0.841\,T_{\text{eff}}$	**boundary temperature**; the atmosphere does not cool to zero
$\tau \gg 1$	$T^4 \approx \frac{3}{4}T_{\text{eff}}^4\tau$	diffusion regime, $F = \frac{4\pi}{3}\frac{dB}{d\tau}$ ✓

9.3 Emergent intensity and limb darkening

Since $S(\tau) = \frac{3F}{4\pi}\left(\tau + \frac{2}{3}\right)$ is **linear**, Eddington–Barbier is exact:

$$I(0,\mu) = S(\tau = \mu) = \frac{3F}{4\pi}\left(\mu + \frac{2}{3}\right) \qquad \Rightarrow \qquad \boxed{\frac{I(0,\mu)}{I(0,1)} = \frac{2 + 3\mu}{5} = \frac{2}{5}\left(1 + \frac{3}{2}\mu\right)}$$

Self-consistency check. The flux computed from the emergent intensity is

$$2\pi\int_0^1 I(0,\mu)\,\mu\,d\mu = 2\pi\cdot\frac{3F}{4\pi}\left(\frac{1}{3} + \frac{1}{3}\right) = F. \checkmark$$

9.4 The exact solution (Milne, Hopf, Chandrasekhar)

Without the Eddington approximation, Step 1 combined with the Schwarzschild–Milne equation (Lecture 7) gives **Milne's integral equation**

$$J(\tau) = \Lambda_\tau[J] = \frac{1}{2}\int_0^\infty J(t)\, E_1|t-\tau|\, dt.$$

Its solution, normalised to flux F, is written (Chandrasekhar eq. 45)

$$\boxed{J(\tau) = \frac{3F}{4\pi}\left[\tau + q(\tau)\right], \qquad T^4 = \frac{3}{4}T_{\text{eff}}^4\left[\tau + q(\tau)\right]}$$

where $q(\tau)$ is the **Hopf function**:

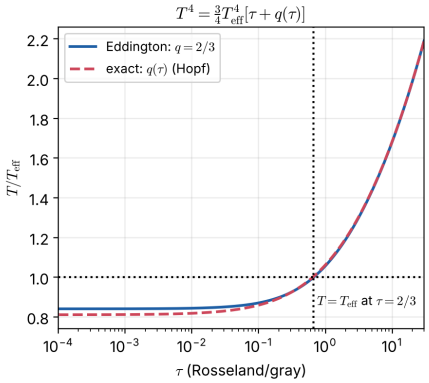
τ	**0**	**0.01**	**0.1**	**0.5**	**1**	**2**	∞
$q(\tau)$ (our solver)	0.5773	0.5882	0.6279	0.6802	0.6985	0.7078	0.7104
exact	$1/\sqrt{3} = 0.57735$						0.710446

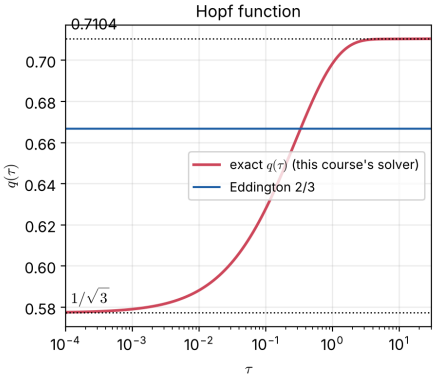
- $q(0) = 1/\sqrt{3}$ is the **Hopf–Bronstein relation** $J(0) = \sqrt{3}\,H$ (Chandrasekhar eq. 64), an exact result.
- Exact boundary temperature: $T(0) = (\sqrt{3}/4)^{1/4}T_{\text{eff}} = 0.8114\,T_{\text{eff}}$, against Eddington's 0.8409 (a 3.6 % error).
- $q(\infty) = 0.7104$ against Eddington's $2/3$: the deep structure is shifted by $\Delta\tau = 0.044$.

Exact emergent intensity (Chandrasekhar eq. 61): $I(0,\mu) = \dfrac{\sqrt{3}}{4}\dfrac{F}{\pi}H(\mu)$, where $H(\mu)$ solves

$$\frac{1}{H(\mu)} = \int_0^1 \frac{\mu' H(\mu')}{2(\mu+\mu')}d\mu', \qquad H(0) = 1,\ H(1) = 2.9078.$$

μ	**0**	**0.1**	**0.2**	**0.5**	**1**
exact $I(\mu)/I(1)$ (Chandrasekhar Table; our solver)	0.3439	0.4290	0.4988	0.6922	1
Eddington $(2+3\mu)/5$	0.400	0.460	0.520	0.700	1





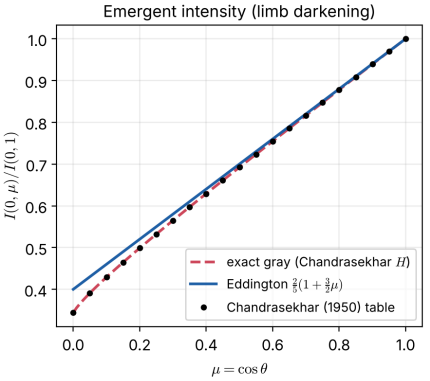


Figure 9.1 — Left: temperature structure, Eddington versus exact. They differ only for $\tau \lesssim 1$. Centre: the Hopf function from our numerical solution of Milne's equation. Right: bolometric limb darkening. The exact H-function law (dashed) agrees with Chandrasekhar's 1950 table (dots) to 4 decimals; the Eddington law (solid) is too bright at the limb.

9.5 Numerical solution of the gray LTE problem

`scripts/lte_solvers.py → hopf_function()` solves Milne's equation directly:

1. Grid τ_i from 0 to 30 (logarithmic).
2. Build the Λ matrix analytically (piecewise-linear S; Lecture 8).
3. Below $\tau_{\max}$, attach the known asymptotic form $S = \frac{3}{4}(\tau + q_\infty)$ and treat its contribution as a known right-hand side.
4. Solve $(\mathbf{1} - \mathbf{\Lambda})\mathbf{S} = \mathbf{b}$; then $q = \frac{4}{3}S - \tau$.

Results: $q(0) = 0.5773$ (exact 0.57735), $q(1) = 0.6985$. The emergent flux, integrated over all wavelengths with $T(\tau)$ from this solution and the LTE formal solution $F_\lambda = 2\pi \int_0^\infty B_\lambda(T(\tau))E_2(\tau)d\tau$, equals $\sigma T_{\rm eff}^4$ to **0.007 %**, confirming radiative equilibrium.

9.6 Monochromatic properties: the spectrum of a gray LTE star

Although κ is gray, the **spectrum is not a black body**, because each frequency samples $B_\nu(T(\tau))$ with T varying with depth:

$$I_\nu(0,\mu) = \int_0^\infty B_\nu\big(T(\tau)\big)\, e^{-\tau/\mu}\,\frac{d\tau}{\mu}, \qquad F_\nu(0) = 2\pi \int_0^\infty B_\nu\big(T(\tau)\big) E_2(\tau)\, d\tau.$$

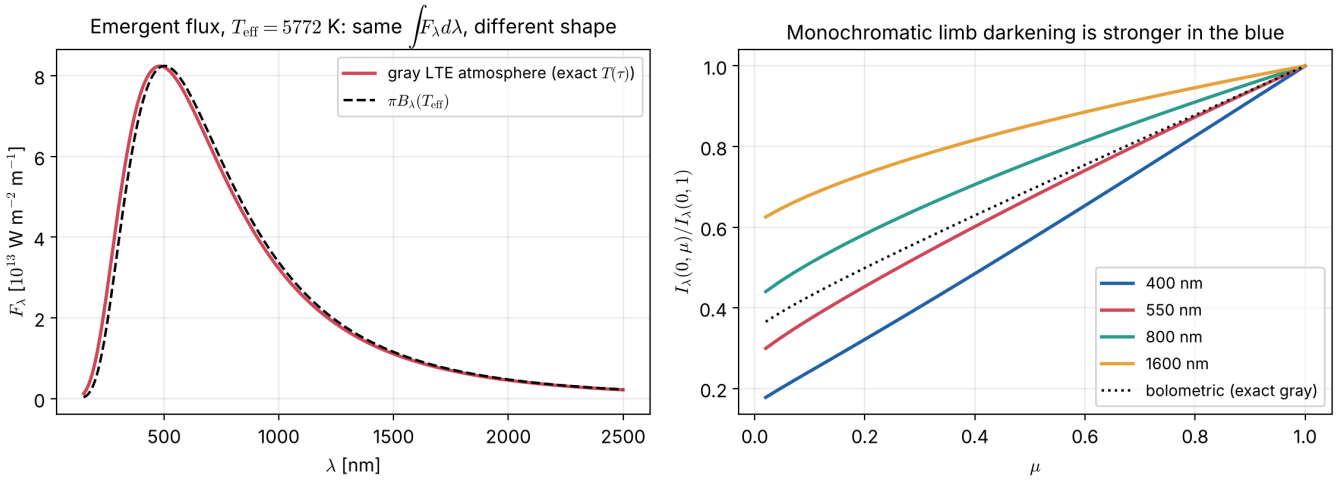


Figure 9.2 — Left: emergent flux of the exact gray LTE atmosphere with $T_{\rm eff} = 5772$ K compared with $\pi B_\lambda(T_{\rm eff})$. The UV excess comes from hot deep layers that shine through. Right: limb darkening is much stronger in the blue, because B_λ is more sensitive to T in the Wien regime (Lecture 3).

λ	$I(0.1)/I(1)$	$I(0.5)/I(1)$
400 nm	0.242	0.569
550 nm	0.372	0.672
800 nm	0.510	0.761

9.7 Adding hydrostatic equilibrium: a complete model

With $dP/dz = -\rho g$ and $d\tau = -\kappa\rho\, dz$:

$$\frac{dP}{d\tau} = \frac{g}{\kappa} \quad \Rightarrow \quad P(\tau) = \frac{g}{\kappa}\tau \quad (\kappa \text{ constant}).$$

Then $\rho = P\mu m_H/(kT)$. Together with $T(\tau)$ this gives a full **gray LTE model atmosphere**: $T(\tau)$, $P(\tau)$, $\rho(\tau)$, as functions of only T_{eff}, g and κ (i.e. composition), exactly as Collins' five assumptions promised (Lecture 6).

Real (non-gray) atmospheres need iteration: guess $T(\tau)$ → Saha–Boltzmann gives κ_ν → solve the RTE at all ν → check $\int \kappa_\nu(J_\nu - B_\nu)d\nu = 0$ and flux constancy → correct T (Unsöld–Lucy correction) → repeat. The gray solution is the standard **starting model**.

Worked example 9.1 — The Sun as a gray LTE atmosphere

$T_{\text{eff}} = 5772$ K. *Find* T *at* $\tau = 0, 2/3, 1, 2, 10$ *(Eddington), and the exact surface temperature.*

τ	0	2/3	1	2	10
T (Eddington) [K]	4854	5772	6103	6864	9707

Exact $T(0) = 0.8114 \times 5772 = 4682$ K. The observed solar temperature minimum is about 4400 K (it then rises in the chromosphere, which a radiative-equilibrium model cannot reproduce). Compare Lecture 7: the limb-darkening inversion at 500 nm gave 5155 K at $\tau_{500} = 0$ and 6300 K at $\tau_{500} = 1$. The gray model is the right order of magnitude, with differences due to non-grayness (H^- opacity varies with λ).

Worked example 9.2 — Pressure at the solar photosphere

Take $g_\odot = 274$ *m s*$^{-2}$ *and a mean opacity* $\kappa \approx 0.03$ *m*^{2} *kg*$^{-1}$. *Estimate* P *and* ρ *at* $\tau = 2/3$.

$P = g\tau/\kappa = 274 \times 0.667/0.03 = 6.1 \times 10^3$ Pa (about 0.06 atm). With $\mu \approx 1.3$ (neutral gas) and $T = 5772$ K: $\rho = P\mu m_H/(kT) = 6.1 \times 10^3 \times 1.3 \times 1.67 \times 10^{-27}/(1.38 \times 10^{-23} \times 5772) = 1.7 \times 10^{-4}$ kg m^{-3}. Detailed solar models give $\rho \approx 2.5 \times 10^{-4}$ kg m^{-3} at $\tau_{500} = 1$, so the gray estimate is good to a factor of order unity.

Worked example 9.3 — Verifying Step 1 directly

Show that if $S = J$ *and* H = const, *then* $\int_0^\infty J(t)E_2(t)dt = 2H$ *(the flux from the Schwarzschild–Milne equation at* $\tau = 0$*), and check it for the Eddington solution.*

$F(0) = 2\pi\int_0^\infty S(t)E_2(t)dt = 4\pi H \Rightarrow \int_0^\infty SE_2\, dt = 2H$. Eddington: $S = 3H(t + 2/3)$, and $\int_0^\infty tE_2 dt = 1/3$, $\int_0^\infty E_2 dt = E_3(0) = 1/2$. So $\int SE_2 = 3H(1/3 + 1/3) = 2H$ ✓. The Eddington solution satisfies the **surface** flux condition exactly, but not Milne's equation at every depth. The exact $q(\tau)$ corrects that.

Exam questions and answers — Lecture 9

Q1. State the assumptions of the gray LTE atmosphere and derive $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$. Indicate clearly where the Eddington approximation enters.

> **A1.** Assumptions: plane-parallel, static, LTE ($S = B$), gray ($\kappa_\nu = \kappa$), radiative equilibrium with $F = \sigma T_{\rm eff}^4$. Derivation: (i) RE with gray opacity gives $J = B$; (ii) $dH/d\tau = J - B = 0 \Rightarrow H = F/4\pi$; (iii) $dK/d\tau = H \Rightarrow K = H\tau + c$ (exact). **(iv) Eddington:** $K = J/3 \Rightarrow J = 3H(\tau + C)$. **(v) Eddington boundary condition** $J(0) = 2H$ gives $C = 2/3$. (vi) $J = B = \sigma T^4/\pi$ and $H = \sigma T_{\rm eff}^4/4\pi$ give $T^4 = \frac{3}{4}T_{\rm eff}^4(\tau + \frac{2}{3})$.

Q2. Derive the limb-darkening law $I(\mu)/I(1) = \frac{2}{5}(1 + \frac{3}{2}\mu)$ and verify that it gives the correct flux.

> **A2.** $S = 3H(\tau + 2/3)$ is linear, so $I(0,\mu) = S(\mu) = 3H(\mu + 2/3)$, and $I(\mu)/I(1) = (\mu + 2/3)/(5/3) = (2 + 3\mu)/5$. Flux: $2\pi \int_0^1 3H(\mu + \frac{2}{3})\mu\, d\mu = 6\pi H(\frac{1}{3} + \frac{1}{3}) = 4\pi H = F$ ✓.

Q3. What is the Hopf function? Give $q(0)$ and $q(\infty)$, and explain how the Eddington solution approximates it.

> **A3.** The exact solution of Milne's equation is $J = 3H[\tau + q(\tau)]$, where q increases monotonically from $q(0) = 1/\sqrt{3} = 0.577$ (Hopf–Bronstein) to $q(\infty) = 0.7104$. Eddington replaces it by the constant $2/3 = 0.667$, which lies between the two limits; the maximum error in T is 3.6 %, at the surface.

Q4. Explain why the emergent spectrum of a gray LTE atmosphere is not a black body at $T_{\rm eff}$, and why it shows an excess in the UV.

> **A4.** $F_\nu = 2\pi \int B_\nu(T(\tau))E_2(\tau)d\tau$ is a weighted average of Planck functions over a range of temperatures. Because B_ν is a convex, steep function of T in the Wien (UV) regime, the hotter layers contribute disproportionately there, so the UV flux exceeds $\pi B_\nu(T_{\rm eff})$. Energy conservation then requires a slight deficit elsewhere. The total $\int F_\nu d\nu$ equals $\sigma T_{\rm eff}^4$ exactly.

Lecture 10 — Spectral Lines in LTE, and the Limits of LTE

> **Sources:** Collins §9.1c (continuum and spectral lines), §9.2e (redistribution: coherent, complete, isotropic); Chandrasekhar Ch. XII §83–84 (line formation; Milne–Eddington model, residual intensities); MIT 8.901 Lec. 7 §7.3 (two-layer atmosphere, thermal inversions); Hu Set 1 ("photons escape through the channel of lowest opacity, so lines are dark"); Lectures 4, 6, 7, 9 of this course.

Learning goals

1. Write the line absorption coefficient, the line profile (Doppler, Lorentz, Voigt) and the Doppler width.
2. Explain with Eddington–Barbier why LTE atmospheres with outward-decreasing T show absorption lines.
3. Derive the Milne–Eddington residual-flux profile exactly in LTE.
4. Explain the curve of growth and use the weak-line formula to measure abundances.
5. Recognise when LTE fails for lines, and what NLTE changes.

10.1 Line opacity

For a bound–bound transition $l \to u$ with oscillator strength f_{lu}:

$$\alpha_\nu^{\text{line}} = \frac{\pi e^2}{4\pi\varepsilon_0 m_e c}\, f_{lu}\, n_l\, \phi(\nu)\left(1 - e^{-h\nu/kT}\right) = \pi r_e c\, f_{lu}\, n_l\, \phi(\nu)\left(1 - e^{-h\nu/kT}\right) \quad [\text{LTE}],$$

with $r_e = 2.818 \times 10^{-15}$ m, so $\pi r_e c = 2.654 \times 10^{-6}$ m^2 Hz. The level population n_l comes from **Saha–Boltzmann** at the local T and P_e (Lecture 6). The factor $\left(1 - e^{-h\nu/kT}\right)$ is the LTE stimulated-emission correction.

Profiles ($\int \phi\, d\nu = 1$):

Broadening	Profile	Width
thermal + microturbulent Doppler	Gaussian: $\phi = \dfrac{1}{\sqrt{\pi}\Delta\nu_D} e^{-v^2}$, $v = (\nu - \nu_0)/\Delta\nu_D$	$\Delta\nu_D = \dfrac{\nu_0}{c}\sqrt{\dfrac{2kT}{m} + \xi^2}$
natural, collisional (pressure), Stark	Lorentzian: $\phi = \dfrac{\gamma/4\pi^2}{(\nu - \nu_0)^2 + (\gamma/4\pi)^2}$	$\gamma = \gamma_{\text{rad}} + \gamma_{\text{coll}}$
both	**Voigt**: $\phi = \dfrac{H(a, v)}{\sqrt{\pi}\Delta\nu_D}$, $a = \dfrac{\gamma}{4\pi\Delta\nu_D}$	Gaussian core + Lorentzian ("damping") wings

10.2 Why lines are dark: LTE and Eddington–Barbier

In LTE with pure absorption, both the continuum and the line emit thermally:

$$S_\nu = \frac{\kappa_c B_\nu + \kappa_l B_\nu}{\kappa_c + \kappa_l} = B_\nu(T).$$

The difference lies only in the **depth scale**. With $\eta_\nu \equiv \kappa_l \phi / \kappa_c$, $d\tau_\nu = (1+\eta_\nu) d\tau_c$. By Eddington–Barbier, $F_\nu \approx \pi B_\nu\big(T(\tau_\nu = 2/3)\big)$:

- In the **continuum**, $\tau_\nu = 2/3$ at $\tau_c = 2/3$: deep and hot.
- At **line centre**, $\tau_\nu = 2/3$ at $\tau_c = \frac{2/3}{1+\eta}$: high and cool.

So $F_{\text{line}} < F_{\text{cont}}$ when T decreases outward. This is the continuous version of MIT 8.901's two-layer toy star (Lecture 5, §5.4), where the continuum shows $B_\nu(T_H)$ and the lines show $B_\nu(T_C)$. **An LTE absorption line is a map of the temperature gradient.** If T increases outward (a chromosphere), the same physics gives emission lines.

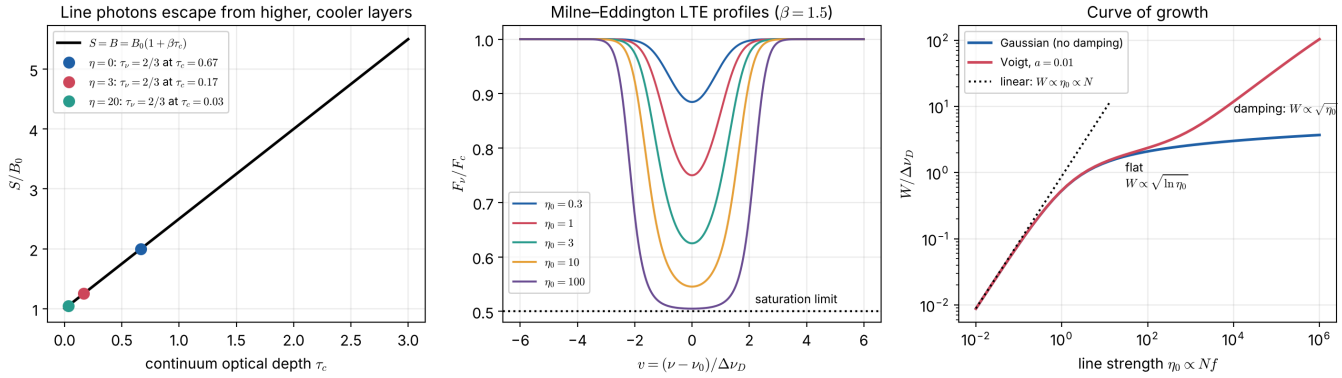


Figure 10.1 — Left: line photons escape from higher layers, where $S = B$ is smaller. Centre: Milne–Eddington LTE profiles for increasing line strength η_0. The core saturates at a finite depth (dotted line), set by the surface value of B. Right: the curve of growth. The equivalent width grows linearly, then flattens (saturation), then rises again as $\sqrt{\eta_0}$ when damping wings become optically thick.

10.3 The Milne–Eddington model: an exact LTE line solution

Assumptions: LTE ($S_\nu = B_\nu$); B_ν linear in continuum optical depth, $B_\nu = B_0(1+\beta\tau_c)$; and η_ν = constant with depth (the same ratio of line to continuum opacity at all depths).

Then $\tau_\nu = (1+\eta_\nu)\tau_c$, so $B_\nu = B_0\left(1 + \dfrac{\beta}{1+\eta_\nu}\tau_\nu\right)$ is linear in τ_ν, and Eddington–Barbier is **exact**:

$$F_\nu = \pi B_0\left(1 + \frac{2}{3}\frac{\beta}{1+\eta_\nu}\right), \qquad F_c = \pi B_0\left(1 + \frac{2\beta}{3}\right),$$

$$\boxed{r_\nu \equiv \frac{F_\nu}{F_c} = \frac{1 + \dfrac{2\beta}{3(1+\eta_\nu)}}{1 + \dfrac{2\beta}{3}}}$$

- Weak line ($\eta \ll 1$): $1 - r_\nu \approx \dfrac{2\beta/3}{1+2\beta/3}\,\eta_\nu \propto \eta_\nu \propto Nf$.

- Saturated core ($\eta \to \infty$): $r \to \dfrac{1}{1+2\beta/3}$. The maximum depth is set by the temperature gradient β. A steeper gradient gives deeper lines; an isothermal atmosphere ($\beta = 0$) gives **no lines at all** in LTE.
- For the gray atmosphere, bolometrically $B \propto \tau + 2/3$, i.e. $\beta = 3/2$, and the maximum depth is $1 - 1/2 = 0.5$. At optical wavelengths β is larger (Wien sensitivity), so lines can be deeper.

10.4 Equivalent width and the curve of growth

The **equivalent width** is the width of a rectangle of zero flux with the same area as the line:

$$W_\nu = \int \left(1 - \frac{F_\nu}{F_c}\right) d\nu, \qquad W_\lambda = \frac{\lambda^2}{c} W_\nu.$$

Weak-line (optically thin) limit. For an absorbing column N_l in front of a continuum (the slab of Lecture 5 with $S \ll I_0$), $1 - F/F_c \approx \tau_\nu$, so

$$\boxed{W_\lambda = \pi r_e \, \lambda^2 \, N_l f} \qquad \text{(independent of the profile and of the broadening)}$$

In SI: $W_\lambda[\mathrm{m}] = 8.853 \times 10^{-15} \, \lambda^2[\mathrm{m}^2] \, N_l f \, [\mathrm{m}^{-2}]$.

Three regimes:

Regime	Condition	W depends on	Use
linear	$\tau_0 \lesssim 1$	$W \propto N f$	**abundances** (best)
flat (saturated)	$1 \lesssim \tau_0 \lesssim 10^{3\text{–}4}$	$W \propto \Delta\nu_D \sqrt{\ln \tau_0}$	**microturbulence** ξ, temperature
damping (square-root)	wings opaque	$W \propto \sqrt{N f \gamma}$	gravity (pressure broadening)

10.5 Beyond LTE

From Lecture 6, the line source function of a two-level atom is

$$S^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B, \qquad \varepsilon \approx \frac{C_{21}}{C_{21} + A_{21}}.$$

For strong lines in low-density layers, $\varepsilon \ll 1$. Then, from Lecture 8:

- $S^{\text{line}}(0) \approx \sqrt{\varepsilon}\, B$, well below B, so **NLTE line cores are darker than LTE cores** (e.g. Na I D, Mg I b, Ca II H & K, Hα cores in the Sun).
- Thermalisation occurs only at $\tau_{\text{line}} \sim 1/\sqrt{\varepsilon}$, which can be deep in the photosphere.

Departure coefficients $b_i = n_i / n_i^{\text{LTE}}$ give

$$S^{\text{line}} = \frac{2h\nu^3}{c^2} \frac{1}{\dfrac{b_l}{b_u} e^{h\nu/kT} - 1}, \qquad S^{\text{line}} \approx \frac{b_u}{b_l} B_\nu \quad \text{(Wien)}.$$

When to expect NLTE: low density (giants, supergiants, upper atmospheres); hot stars with intense UV radiation (photo-ionisation dominates); low metallicity (fewer free electrons, so fewer collisions); strong resonance lines; nebulae

and the ISM (always NLTE).

	LTE	NLTE
populations	Saha–Boltzmann(T, n_e)	rate equations, coupled to J_ν
S^{line}	$B_\nu(T)$	$(1-\varepsilon)\bar{J} + \varepsilon B$, or via b_i
locality	local	non-local (photons couple distant layers)
computational cost	small	large (ALI, multi-level, many frequencies)
where valid	deep photospheres, weak lines, high density	strong lines, chromospheres, hot or low-density stars

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Worked example 10.1 — Doppler widths

Compute the thermal Doppler width for (a) Hα (656.28 nm) at 5772 K, (b) an Fe I line at 500 nm at 5772 K.

a. $v_{\text{th}} = \sqrt{2kT/m_H} = 9.76$ km s^{-1}, so $\Delta\lambda_D = \lambda v/c = 0.0214$ nm $= 214$ mÅ.
b. $m = 55.85\, m_u$: $v_{\text{th}} = 1.31$ km s^{-1}, so $\Delta\lambda_D = 21.9$ mÅ. Iron lines are about 10 times narrower than hydrogen lines. In the Sun the microturbulence $\xi \approx 1$ km s^{-1} is comparable to v_{th}(Fe), so it matters for metal lines.

Worked example 10.2 — An abundance from a weak line

A weak Fe I line at 500 nm has $W_\lambda = 22$ mÅ and $f = 0.1$. Assume it is on the linear part of the curve of growth and treat the atmosphere as an absorbing column. Find the column density N_l of absorbing atoms.

$$N_l = \frac{W_\lambda}{\pi r_e \lambda^2 f} = \frac{2.2\times10^{-12}}{8.853\times10^{-15}\times(5\times10^{-7})^2\times 0.1} = 1.0\times10^{16}\ \text{m}^{-2}.$$

The total iron abundance follows from Saha–Boltzmann (fraction of Fe in the lower level of Fe I) and the column of hydrogen above $\tau \approx 2/3$, roughly $P/(g\mu m_H)$ from Lecture 9. **LTE enters at the Saha–Boltzmann step.**

Worked example 10.3 — Milne–Eddington line depths

With $\beta = 1.5$, find the central residual flux for $\eta_0 = 1$, 10 and ∞.

$r = \dfrac{1 + 1/(1+\eta)}{2}$: for $\eta = 1$, $r = 0.75$; for $\eta = 10$, $r = 0.545$; for $\eta \to \infty$, $r = 0.5$. Increasing η from 10 to ∞ barely changes the core: the line is **saturated**, and further growth of W happens only in the wings.

Worked example 10.4 — LTE emission lines from a chromosphere

Suppose that above the temperature minimum T rises outward, so that near the surface $B = B_0(1 - \beta'\tau_c)$ with $\beta' > 0$ for $\tau_c < 0.1$. What does LTE predict for a strong line formed at $\tau_c \approx 0.01$?

The strong line's $\tau_\nu = 2/3$ falls at $\tau_c \approx 0.01$, where B is larger than at $\tau_c \approx 0.1$ (the temperature minimum), so the core shows an **emission reversal** inside the broad absorption profile. This is seen in the Ca II H & K lines (the K_2

peaks). In reality NLTE decoupling ($S < B$) makes the emission weaker than LTE predicts.

Exam questions and answers — Lecture 10

Q1. Explain, using the Eddington–Barbier relation, why the solar spectrum shows absorption lines. What would an LTE isothermal atmosphere show?

> **A1.** $F_\nu \approx \pi S_\nu(\tau_\nu = 2/3)$, and in LTE $S_\nu = B_\nu(T)$. Line opacity raises τ_ν, so $\tau_\nu = 2/3$ is reached higher up, where T (hence B_ν) is lower, and the flux is lower. If the atmosphere were isothermal, B_ν would be the same at all depths and $F_\nu = F_c$: no lines, whatever the opacity.

Q2. Derive the Milne–Eddington residual flux and its saturation limit.

> **A2.** See §10.3. $B = B_0(1 + \beta\tau_c)$ and $\tau_\nu = (1+\eta)\tau_c$ give $B = B_0(1 + \beta\tau_\nu/(1+\eta))$, which is linear in τ_ν. Then $F = \pi B(\tau_\nu = 2/3) = \pi B_0(1 + \frac{2\beta}{3(1+\eta)})$. Divide by F_c ($\eta = 0$). As $\eta \to \infty$, $r \to 1/(1 + 2\beta/3)$.

Q3. Sketch the curve of growth and explain the physics of each of its three parts. Which part is best for abundance determination and why?

> **A3.** Linear: the line is optically thin; every atom removes photons independently, so $W \propto Nf$. Flat: the core is saturated ($r \to r_{\min}$), and W grows only as the Doppler core broadens in frequency where $\tau_\nu > 1$, i.e. $W \propto \Delta\nu_D\sqrt{\ln \tau_0}$. Square-root: the Lorentzian damping wings ($\propto \gamma/\Delta\nu^2$) become optically thick, giving $W \propto \sqrt{Nf\gamma}$. The linear part is best for abundances because W is insensitive to the poorly known broadening parameters (ξ, γ).

Q4. Why are the cores of strong resonance lines in the Sun deeper than LTE models predict?

> **A4.** In the low-density upper photosphere and chromosphere, $C_{21} \ll A_{21}$, so $\varepsilon \ll 1$ and $S^{\text{line}} = (1-\varepsilon)\bar{J} + \varepsilon B$. Photons escape from the line core ($\bar{J} < B$) and the source function drops to $\sim \sqrt{\varepsilon}\, B$ at the surface. The core intensity $\approx S(\tau_\nu \approx 1)$ is therefore lower than $B(T)$, and the line is darker than in LTE.

Final Examination — Radiative Transfer (with complete solutions)

> **Format suggestion:** 3 hours. Part A (30 %): short answers. Part B (70 %): problems; answer any 8 of the 12. The solutions follow each question. All numbers were checked with `scripts/worked_examples.py` and the solvers in `scripts/lte_solvers.py`. Constants: $h = 6.626 \times 10^{-34}$ J s, $k = 1.381 \times 10^{-23}$ J K^{-1}, $c = 2.998 \times 10^{8}$ m s^{-1}, $\sigma = 5.670 \times 10^{-8}$ W m^{-2} K^{-4}, $m_e = 9.109 \times 10^{-31}$ kg, $m_H = 1.674 \times 10^{-27}$ kg, $r_e = 2.818 \times 10^{-15}$ m, $\sigma_T = 6.652 \times 10^{-29}$ m^2, $R_\odot = 6.957 \times 10^{8}$ m, 1 pc $= 3.086 \times 10^{16}$ m, 1 eV $= 1.602 \times 10^{-19}$ J. $\int_0^\infty E_n(t)\,dt = \dfrac{1}{n}$, $\int_0^\infty t\,E_n(t)\,dt = \dfrac{1}{n+1}$, $E_n(0) = \dfrac{1}{n-1}$.

Part A — Short answers (2 marks each)

A1. True or false: “In LTE the specific intensity equals the Planck function.” > **False.** In LTE the **source function** equals $B_\nu(T)$. $I_\nu = B_\nu$ only in full TE, or deep inside an optically thick, isothermal region.

A2. What is conserved along a ray in vacuum: F_ν, I_ν or J_ν? > I_ν (more generally I_ν/ν^3).

A3. Give the source function for coherent isotropic scattering plus thermal absorption. > $S_\nu = \varepsilon B_\nu + (1-\varepsilon) J_\nu$, with $\varepsilon = \kappa^a/(\kappa^a + \sigma^{\rm sc})$.

A4. At what optical depth does the emergent flux “originate” according to Eddington–Barbier? > $\tau_\nu = 2/3$: $F_\nu(0) \approx \pi S_\nu(2/3)$.

A5. Why is the Rosseland mean a harmonic mean? > The flux in each frequency band is $\propto 1/\kappa_\nu$ (like conductances in parallel), so the total is dominated by low-opacity windows.

A6. Write the Eddington closure and the Eddington surface condition. > $K = J/3$; $J(0) = 2H(0)$.

A7. What does the factor $1 - e^{-h\nu/kT}$ represent? > The LTE correction for stimulated emission (negative absorption).

A8. For an isothermal scattering atmosphere, what is $S(0)/B$ exactly? > $\sqrt{\varepsilon}$.

A9. What is the exact boundary temperature of a gray atmosphere in units of $T_{\rm eff}$? > $(\sqrt{3}/4)^{1/4} = 0.811$.

A10. Why do Balmer lines peak in A stars? > Boltzmann excitation into $n=2$ increases with T, while Saha ionisation removes neutral H above about 10^4 K. The product peaks near 9900 K.

A11. What is an optically thin source’s brightness temperature in terms of T and τ? > $T_b = T(1 - e^{-\tau}) \approx T\tau$ (RJ, no background).

A12. Name the integral equation of the gray LTE atmosphere. > Milne’s equation, $J(\tau) = \Lambda_\tau[J] = \frac{1}{2}\int_0^\infty J(t) E_1|t-\tau|\,dt$.

A13. Two stars with equal $T_{\rm eff}$ but different radii: which quantity is the same, the flux at Earth or the surface brightness? > The surface brightness.

A14. Why are NLTE effects strongest in strong resonance lines? > They form in high, low-density layers where $C \ll A$ ($\varepsilon \ll 1$), and the thermalisation depth $1/\sqrt{\varepsilon}$ is very large.

A15. What does a positive β in $B = B_0(1 + \beta\tau)$ imply for limb darkening and for line formation in LTE? > Temperature increasing inward: limb darkening, and absorption lines.

Part B — Problems

B1. Intensity and flux (8 marks)

An A-type star has $R = 1.7 R_\odot$ and $T_{\rm eff} = 9900$ K and lies at $d = 2.64$ pc. (a) Find its surface flux and luminosity in $L_\odot$ ($L_\odot = 3.828 \times 10^{26}$ W). (b) Find the bolometric flux at Earth. (c) Show that its mean surface brightness is the same at the star and at Earth.

> **Solution.** (a) $F_s = \sigma T^4 = 5.67 \times 10^{-8} \times (9900)^4 = 5.45 \times 10^8$ W m^{-2}; $L = 4\pi R^2 F_s = 25\, L_\odot$. (b) $F_\oplus = F_s (R/d)^2$, with $R = 1.18 \times 10^9$ m and $d = 8.15 \times 10^{16}$ m, so $(R/d)^2 = 2.1 \times 10^{-16}$ and $F_\oplus = 1.15 \times 10^{-7}$ W m^{-2}. (c) The mean intensity is $\bar I = F_s/\pi$. At Earth, $\bar I = F_\oplus / \Omega$ with $\Omega = \pi (R/d)^2$, giving $F_s (R/d)^2 / (\pi (R/d)^2) = F_s/\pi$, the same. Intensity is conserved along rays.

B2. Planck function (8 marks)

a. Show that in the RJ limit $I_\nu = (2k\nu^2/c^2) T_b$. (b) A compact source has $I_\nu = 10^{-18}$ W m^{-2} Hz^{-1} sr^{-1} at 5 GHz. Find T_b. (c) If the source is an H II region with $T_e = 10^4$ K, what is its optical depth at 5 GHz? (d) Is it thick or thin at 0.5 GHz if $\tau \propto \nu^{-2.1}$?

> **Solution.** (a) $e^x - 1 \approx x$ gives $B_\nu \approx 2h\nu^3/(c^2 x) = 2\nu^2 kT/c^2$; define T_b by $I_\nu = B_\nu(T_b)$. (b) $T_b = c^2 I_\nu / (2k\nu^2) = (8.99 \times 10^{16} \times 10^{-18})/(2 \times 1.381 \times 10^{-23} \times 2.5 \times 10^{19}) = 130$ K. (c) $T_b = T_e(1 - e^{-\tau}) \Rightarrow 1 - e^{-\tau} = 0.013 \Rightarrow \tau = 0.013$. (d) $\tau(0.5\ \text{GHz}) = 0.013 \times 10^{2.1} = 1.6$: marginally thick, so $T_b = 10^4 (1 - e^{-1.6}) = 8000$ K.

B3. Kirchhoff's law and LTE (8 marks)

a. Using the Einstein coefficients, write j_ν and α_ν for a line. (b) Show that if the level populations obey Boltzmann's law at temperature T, then $S_\nu = B_\nu(T)$. (c) Explain why this result does not require $I_\nu = B_\nu$.

> **Solution.** (a) $j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi$, $\alpha_\nu = \frac{h\nu}{4\pi}\phi (n_1 B_{12} - n_2 B_{21})$. (b) $S = \frac{n_2 A_{21}}{n_1 B_{12} - n_2 B_{21}} = \frac{A_{21}/B_{21}}{\frac{n_1 g_2}{n_2 g_1} - 1}$ (using $g_1 B_{12} = g_2 B_{21}$). Boltzmann gives $\frac{n_1 g_2}{n_2 g_1} = e^{h\nu/kT}$, and $A_{21}/B_{21} = 2h\nu^3/c^2$, so $S = B_\nu(T)$. (c) Only the **populations** (a property of the matter, set by collisions) were used. The radiation field does not appear, so $S = B$ holds whenever the matter is in LTE, whatever I_ν is.

B4. Saha equation (8 marks)

Pure hydrogen has $n_e = 10^{20}$ m^{-3}. Find the ionised fraction at (a) $T = 6000$ K and (b) $T = 8000$ K. ($U_{\rm I} = 2$, $U_{\rm II} = 1$, $\chi = 13.6$ eV.)

> **Solution.** $\frac{N_{\rm II}}{N_{\rm I}} = \frac{1}{n_e}\left(\frac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi/kT}$ (the factor $2U_{\rm II}/U_{\rm I} = 1$). (a) $T = 6000$ K: $(\dots)^{3/2} = 2.415\times10^{21}\times 6000^{1.5} = 1.12\times10^{27}$ m^{-3}; $\chi/kT = 26.3$, so $e^{-26.3} = 3.8\times10^{-12}$. Ratio $= 1.12\times10^{27}\times3.8\times10^{-12}/10^{20} = 4.2\times10^{-5}$: essentially neutral. (b) $T = 8000$ K: $(\dots)^{3/2} = 1.73\times10^{27}$; $\chi/kT = 19.7$, so $e^{-19.7} = 2.7\times10^{-9}$. Ratio $= 0.047$; ionised fraction $= 0.045$. A 33 % rise in T increases ionisation by a factor of about 1000: Saha ionisation is extremely sensitive to temperature.

B5. Formal solution in a plane-parallel atmosphere (8 marks)

$S(\tau) = a + b\tau + c\tau^2$ in a semi-infinite atmosphere. (a) Find $I(0,\mu)$. (b) Find $F(0)$. (c) Compare with the Eddington–Barbier values and state when they are exact.

> **Solution.** (a) $I(0,\mu) = \int_0^\infty (a + bt + ct^2)e^{-t/\mu}dt/\mu = a + b\mu + 2c\mu^2$. (b) $F = 2\pi\int_0^1 (a + b\mu + 2c\mu^2)\mu\,d\mu = 2\pi(a/2 + b/3 + c/2) = \pi(a + \tfrac{2}{3}b + c)$. (c) E–B: $I \approx S(\mu) = a + b\mu + c\mu^2$ (error $c\mu^2$); $F \approx \pi S(2/3) = \pi(a + \tfrac{2}{3}b + \tfrac{4}{9}c)$ (error $\tfrac{5}{9}\pi c$). Both are exact when $c = 0$ (linear S).

B6. The gray LTE atmosphere — full derivation (12 marks)

a. Starting from the plane-parallel RTE with $S = B$ and gray opacity, show that radiative equilibrium implies $J = B$ and H = constant. (b) Using the Eddington approximation, derive $T(\tau)$. (c) For $T_{\rm eff} = 10\,000$ K, tabulate T at $\tau = 0, 2/3, 1, 4$. (d) Derive the emergent intensity $I(0,\mu)$ and the limb-darkening law. (e) The exact surface temperature is $0.811\,T_{\rm eff}$. What is the percentage error of the Eddington value?

> **Solution.** (a) RE: $\int \kappa_\nu(J_\nu - B_\nu)d\nu = 0$, which with $\kappa_\nu = \kappa$ gives $J = B$. The zeroth moment $dH/d\tau = J - S = J - B = 0$ gives H = const $= \sigma T_{\rm eff}^4/4\pi$. (b) $dK/d\tau = H \Rightarrow K = H\tau +$ const; $K = J/3 \Rightarrow J = 3H(\tau + C)$; $J(0) = 2H \Rightarrow C = 2/3$. $J = B = \sigma T^4/\pi$ gives $T^4 = \tfrac{3}{4}T_{\rm eff}^4(\tau + \tfrac{2}{3})$. (c) $T(0) = 8409$ K, $T(2/3) = 10\,000$ K, $T(1) = 10\,574$ K, $T(4) = 13\,678$ K. (d) $S = 3H(\tau + 2/3)$ is linear, so $I(0,\mu) = 3H(\mu + 2/3) = \frac{3\sigma T_{\rm eff}^4}{4\pi}(\mu + \tfrac{2}{3})$, and $I(\mu)/I(1) = (2 + 3\mu)/5$. (e) $(0.841 - 0.811)/0.811 = 3.7\,\%$.

B7. Testing the Eddington solution with the Λ operator (8 marks)

For the Eddington gray solution $S(\tau) = 3H(\tau + \tfrac{2}{3})$, compute $J(0) = \Lambda_0[S] = \tfrac{1}{2}\int_0^\infty S(t)E_1(t)dt$. Compare with $S(0)$ and with the exact (Hopf–Bronstein) value $J(0) = \sqrt{3}H$. What does this tell you?

> **Solution.** $J(0) = \frac{3H}{2}\left[\int_0^\infty tE_1 dt + \frac{2}{3}\int_0^\infty E_1 dt\right] = \frac{3H}{2}\left[\frac{1}{2} + \frac{2}{3}\right] = \frac{7}{4}H = 1.750\,H$. But $S(0) = 2H$, so $\Lambda[S] \neq S$: the Eddington solution does **not** satisfy Milne's equation exactly ($J \neq S$ at the surface).

The exact value $\sqrt{3}H = 1.732H$ is close to the result of one Λ-iteration (1.750H), which shows that Λ-iteration improves the solution near the surface, where it couples to the boundary.

B8. Two-stream (Schuster–Schwarzschild) gray atmosphere (10 marks)

Represent the radiation field by two streams, I^+ at $\mu = +1/\sqrt{3}$ and I^- at $\mu = -1/\sqrt{3}$, so that $J = \tfrac{1}{2}(I^+ + I^-)$, $H = \tfrac{1}{2\sqrt{3}}(I^+ - I^-)$, $K = \tfrac{1}{6}(I^+ + I^-)$. (a) Show that $K = J/3$ exactly. (b) With $I^-(0) = 0$, show $J(0) = \sqrt{3}H$. (c) Hence derive $T^4 = \tfrac{3}{4}T_{\rm eff}^4(\tau + 1/\sqrt{3})$ and compare $T(0)$ with the exact result.

Solution. (a) $K = \tfrac{1}{2}\sum \mu_i^2 I_i = \tfrac{1}{2}\cdot\tfrac{1}{3}(I^+ + I^-) = J/3$. (b) $J(0) = I^+/2$ and $H(0) = I^+/(2\sqrt{3})$, so $J(0) = \sqrt{3}H(0)$. (c) As in B6: $J = 3H(\tau + C)$ with $3HC = \sqrt{3}H \Rightarrow C = 1/\sqrt{3}$. So $T^4 = \tfrac{3}{4}T_{\rm eff}^4(\tau + 0.577)$ and $T(0) = (\sqrt{3}/4)^{1/4}T_{\rm eff} = 0.811\,T_{\rm eff}$, which is **exactly the true surface value**. This is Chandrasekhar's "first approximation" of the discrete-ordinate method; it gives $J(0)$ exactly in every order (Chandrasekhar eq. 64), while $q(\infty)$ improves with the order.

B9. Limb darkening and LTE temperature structure (8 marks)

At $\lambda = 500$ nm, the observed solar limb darkening is $I(\mu) = I(1)(0.36 + 0.64\mu)$ with $I(1) = 4.0 \times 10^{13}$ W m^{-3} sr^{-1}. Assuming LTE and the Eddington–Barbier relation, find $T(\tau_{500} = 0)$ and $T(\tau_{500} = 1)$. Comment.

Solution. $S(\tau) = I(1)(0.36 + 0.64\tau)$, and LTE gives $S = B_\lambda(T)$, so $T = (hc/\lambda k)/\ln(1 + 2hc^2/(\lambda^5 B))$, with $hc/\lambda k = 28\,776$ K and $2hc^2/\lambda^5 = 3.811 \times 10^{15}$. $\tau = 0$: $B = 1.44 \times 10^{13}$, so $T = 28776/\ln(265.7) = 5155$ K. $\tau = 1$: $B = 4.0 \times 10^{13}$, so $T = 28776/\ln(96.3) = 6300$ K. The temperature rises inward by about 1100 K per unit optical depth, which is why the Sun shows limb darkening and absorption lines. The values bracket $T_{\rm eff} = 5772$ K, consistent with the gray model (Lecture 9).

B10. Scattering and thermalisation (10 marks)

An isothermal atmosphere has $S = \varepsilon B + (1-\varepsilon)J$. (a) In the Eddington approximation, derive the differential equation for J and solve it with the appropriate boundary conditions. (b) For $\varepsilon = 10^{-2}$, find $S(0)/B$ (Eddington) and compare with the exact value. (c) Estimate the thermalisation depth and the number of scatterings a photon undergoes before destruction. (d) Why does Λ-iteration starting from $S = B$ converge so slowly?

Solution. (a) $dH/d\tau = J - S = \varepsilon(J - B)$ and $H = \tfrac{1}{3}dJ/d\tau$, so $\tfrac{1}{3}J'' = \varepsilon(J - B)$. Then $J = B + Ce^{-\sqrt{3\varepsilon}\tau}$; with $J(0) = \tfrac{2}{3}J'(0)$, $C = -B/(1 + 2\sqrt{\varepsilon/3})$. (b) $\sqrt{\varepsilon/3} = 0.0577$, so $J(0)/B = 1 - 1/1.1155 = 0.1035$ and $S(0)/B = 0.01 + 0.99 \times 0.1035 = 0.1125$. The exact value is $\sqrt{\varepsilon} = 0.100$; Eddington overestimates by 12 %. (c) $\tau_{\rm th} \sim 1/\sqrt{\varepsilon} = 10$ (Eddington: $1/\sqrt{3\varepsilon} = 5.8$); $N \sim 1/\varepsilon = 100$ scatterings. (d) Each iteration transmits information only about $\Delta\tau \sim 1$, and the error decreases by a factor $(1 - \varepsilon) = 0.99$ per step, so hundreds of iterations are needed (see Figure 8.1). Our solver gives $S(0)$ errors of 95 % after 10 iterations, 17 % after 50, 5 % after 100 and 0.9 % after 200; yet beyond 100 iterations each step changes $S(0)$ by less than 0.1 %, so a naive convergence test would stop far too early.

B11. Milne–Eddington line in LTE (10 marks)

In an LTE atmosphere $B_\nu = B_0(1+\beta\tau_c)$, and a line has a constant ratio $\eta_\nu = \eta_0 e^{-v^2}$ of line to continuum opacity. (a) Derive F_ν/F_c. (b) For $\beta = 3$, find the central depth for $\eta_0 = 0.1,\, 1,\, \infty$. (c) Show that for weak lines the equivalent width is proportional to η_0, and find the constant (in units of $\Delta\nu_D$).

> **Solution.** (a) $\tau_\nu = (1+\eta_\nu)\tau_c$, so B is linear in τ_ν and E–B is exact: $r = \frac{1+2\beta/[3(1+\eta)]}{1+2\beta/3}$. (b) $\beta = 3$: $1 + 2\beta/3 = 3$. $\eta_0 = 0.1$: $r = (1+2/1.1)/3 = 0.939$, depth 0.061. $\eta_0 = 1$: $r = 2/3$, depth 0.333. $\eta_0 \to \infty$: $r = 1/3$, depth 0.667. (c) For $\eta \ll 1$: $1 - r \approx \frac{2\beta/3}{1+2\beta/3}\eta_\nu$, so $W/\Delta\nu_D = \int (1-r)dv = \frac{2\beta/3}{1+2\beta/3}\eta_0\sqrt{\pi}$. For $\beta = 3$: $W/\Delta\nu_D = \frac{2}{3}\sqrt{\pi}\,\eta_0 = 1.18\,\eta_0$.

B12. Diffusion and Rosseland mean (8 marks)

a. From $I \approx B + \mu\, dB/d\tau$, derive $F = \frac{4\pi}{3}\frac{dB}{d\tau}$. (b) A gas has three frequency bands carrying 50 %, 30 % and 20 % of $\partial B/\partial T$, with $\kappa = 1,\, 10$ and 1000 (arbitrary units). Find the Rosseland and Planck means. (c) Which mean controls the energy flux in a stellar interior, and why?

> **Solution.** (a) $F = 2\pi\int_{-1}^{1}(B + \mu B')\mu\, d\mu = 2\pi B' \cdot \frac{2}{3} = \frac{4\pi}{3}B'$. (b) $1/\kappa_R = 0.5/1 + 0.3/10 + 0.2/1000 = 0.5302 \Rightarrow \kappa_R = 1.89$. $\kappa_P = 0.5 + 3 + 200 = 203.5$. (c) The Rosseland mean, because the flux is $\propto \sum(1/\kappa_\nu)$: radiation leaks through the most transparent band, so $\kappa_R \approx 2\times$ the window opacity, while the Planck mean is dominated by the most opaque band.

Part C — Essay question (choose one, 10 marks)

C1. "Local thermodynamic equilibrium is a statement about matter, not about radiation." Discuss, with reference to Kirchhoff's law, the two-level atom and the thermalisation depth.

> **Model answer outline.** (1) Define TE and LTE; Maxwell, Boltzmann and Saha at the local T. (2) Kirchhoff: $j = \alpha B$ depends only on the matter state, hence $S = B$ in LTE while $I \neq B$ (escape at the surface; $I(0, \mu < 0) = 0$; $J(0) \approx B/2$). (3) Validity: collisions must dominate radiative rates; two-level atom $S = (1-\varepsilon)\bar{J} + \varepsilon B$, so LTE when $\varepsilon \to 1$. (4) Even with ε close to but below 1, S deviates from B within $\tau \sim 1/\sqrt{\varepsilon}$ of the surface, with $S(0) = \sqrt{\varepsilon}B$. (5) Consequences: LTE is good for continua and weak lines in dwarf photospheres; it fails for strong lines, chromospheres, hot or low-density stars and nebulae. (6) Practical: LTE turns the RTE into an explicit formal solution once $T(\tau)$ is known; NLTE requires solving rate equations coupled to the radiation field (ALI).

C2. Describe how a model stellar atmosphere is computed, starting from the gray LTE solution, and how its predictions are compared with observations.

> **Model answer outline.** Inputs $T_{\rm eff}$, $\log g$, composition → gray starting $T(\tau) = [\frac{3}{4}T_{\rm eff}^4(\tau + q)]^{1/4}$ → hydrostatic $P(\tau)$ from $dP/d\tau = g/\kappa$ → Saha–Boltzmann populations → $\kappa_\nu,\ j_\nu$ → solve the RTE at many

frequencies (formal solution, or Feautrier/ALI with scattering) → check radiative equilibrium $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$ and constant flux → temperature correction (Unsöld–Lucy) → iterate to convergence → emergent spectrum, limb darkening, line profiles (Milne–Eddington intuition, curve of growth) → compare with observed spectra, colours, Balmer jump and line equivalent widths → derive $T_{\rm eff}$, g and abundances.

Formula Sheet — Radiative Transfer (for revision and the exam)

Radiation field (Lecture 2)

Specific intensity	$dE = I_\nu \cos\theta \, dA \, d\Omega \, d\nu \, dt$; I_ν constant along rays in vacuum; I_ν/ν^3 invariant
Mean intensity	$J_\nu = \frac{1}{4\pi}\oint I_\nu d\Omega = \frac{1}{2}\int_{-1}^{1} I_\nu d\mu$
Flux	$F_\nu = \oint I_\nu \cos\theta \, d\Omega = 2\pi \int_{-1}^{1} I_\nu \mu \, d\mu$; $H_\nu = F_\nu/4\pi$
K-integral, pressure	$K_\nu = \frac{1}{2}\int_{-1}^{1} I_\nu \mu^2 d\mu$; $P_\nu = 4\pi K_\nu/c$
Energy density	$u_\nu = 4\pi J_\nu/c$
Isotropic field	$F = 0, K = J/3, P = u/3$
Uniform hemisphere	$F = \pi I$
Sphere seen from d	$F = \pi I (R/d)^2$
$\nu \leftrightarrow \lambda$	$I_\lambda = (c/\lambda^2) I_\nu$

Black body (Lecture 3)

$$B_\nu = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/kT}-1}, \quad B_\lambda = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/\lambda kT}-1}$$

- RJ: $B_\nu \approx 2\nu^2 kT/c^2$ · Wien: $B_\nu \approx \frac{2h\nu^3}{c^2} e^{-h\nu/kT}$ · $h\nu_{\max} = 2.821kT \cdot \lambda_{\max} T = 2.898 \times 10^{-3}$ m K
- $\int B_\nu d\nu = \sigma T^4/\pi \cdot F = \sigma T^4 \cdot u = aT^4, a = 4\sigma/c \cdot dB/dT = 4\sigma T^3/\pi \cdot n_\gamma = 2.03 \times 10^7 T^3$ m^{-3}
- T_b: $I_\nu = B_\nu(T_b)$, RJ $T_b = c^2 I_\nu / 2k\nu^2 \cdot L = 4\pi R^2 \sigma T_{\text{eff}}^4$

Coefficients (Lecture 4)

- $\alpha_\nu = n\sigma_\nu = \rho\kappa_\nu \cdot \ell = 1/\alpha_\nu \cdot \tau_\nu = \int \alpha_\nu ds \cdot S_\nu = j_\nu/\alpha_\nu \cdot \kappa_{\text{es}} = 0.02(1+X)$ m^2 kg^{-1}
- Einstein: $g_1 B_{12} = g_2 B_{21}$, $A_{21} = \frac{2h\nu^3}{c^2} B_{21}$
- $j_\nu = \frac{h\nu}{4\pi} n_2 A_{21} \phi$, $\alpha_\nu = \frac{h\nu}{4\pi}\phi(n_1 B_{12} - n_2 B_{21})$, $S = \frac{2h\nu^3/c^2}{g_2 n_1/g_1 n_2 - 1}$
- Scattering + absorption: $S = \varepsilon B + (1-\varepsilon)J$ · Random walk: $R = \sqrt{N}\ell$, $N \approx \max(\tau, \tau^2)$

Transfer equation (Lectures 5, 7)

- $\frac{dI_\nu}{d\tau_\nu} = S_\nu - I_\nu$ (τ along the ray) · $I = I_0 e^{-\tau} + \int_0^\tau S(t) e^{-(\tau - t)} dt$ · slab: $I = I_0 e^{-\tau} + S(1 - e^{-\tau})$

- Plane-parallel (τ inward): $\mu\dfrac{dI}{d\tau} = I - S \cdot I(0,\mu) = \int_0^\infty S\, e^{-t/\mu} dt/\mu$
- $J = \frac{1}{2}\int_0^\infty S\, E_1|t-\tau| d\tau \cdot F = 2\pi\left[\int_\tau^\infty S E_2(t-\tau)dt - \int_0^\tau S E_2(\tau - t)dt\right] \cdot K = \frac{1}{2}\int S E_3|t-\tau| dt$
- $E_n(x) = \int_1^\infty e^{-xt} t^{-n} dt$, $E_n' = -E_{n-1}$, $E_n(0) = 1/(n-1)$, $\int_0^\infty E_n = 1/n$, $\int_0^\infty t E_n = 1/(n+1)$
- **Eddington–Barbier:** $I(0,\mu) \approx S(\tau = \mu)$, $F(0) \approx \pi S(\tau = 2/3)$ (exact for linear S)

LTE (Lecture 6)

- Kirchhoff: $j_\nu = \alpha_\nu B_\nu(T) \Rightarrow S_\nu = B_\nu(T)$ (but $I_\nu \neq B_\nu$)
- Boltzmann: $\dfrac{N_j}{N_i} = \dfrac{g_j}{g_i} e^{-\Delta E/kT}$ · Saha: $\dfrac{N_{r+1} n_e}{N_r} = \dfrac{2U_{r+1}}{U_r}\left(\dfrac{2\pi m_e kT}{h^2}\right)^{3/2} e^{-\chi/kT}$, $(2\pi m_e k/h^2)^{3/2} = 2.415 \times 10^{21}\ \mathrm{m^{-3} K^{-3/2}}$
- LTE opacity factor $1 - e^{-h\nu/kT}$ · two-level atom: $S = (1-\varepsilon)\bar{J} + \varepsilon B$, $\varepsilon' = \frac{C_{21}}{A_{21}}(1 - e^{-h\nu/kT})$, $\varepsilon = \frac{\varepsilon'}{1+\varepsilon'} \cdot n_{\mathrm{crit}} = A_{21}/q_{21}$

Moments, diffusion, scattering (Lecture 8)

- $\dfrac{dH}{d\tau} = J - S \cdot \dfrac{dK}{d\tau} = H$ · RE: $\int \kappa_\nu (J_\nu - B_\nu) d\nu = 0$, $F = \sigma T_{\mathrm{eff}}^4$ = const
- Eddington: $K = J/3$, $J(0) = 2H(0) \Rightarrow \frac{1}{3}J'' = J - S$
- Diffusion: $F_\nu = \frac{4\pi}{3}\frac{dB_\nu}{d\tau_\nu} \cdot \dfrac{1}{\kappa_R} = \dfrac{\int \kappa_\nu^{-1} \partial_T B_\nu d\nu}{\int \partial_T B_\nu d\nu} \cdot F = -\dfrac{16\sigma T^3}{3\kappa_R \rho}\dfrac{dT}{dz}$
- Isothermal scattering: $S(0) = \sqrt{\varepsilon}\, B$ (exact) · Eddington $J = B\left[1 - \frac{e^{-\sqrt{3\varepsilon}\tau}}{1+2\sqrt{\varepsilon/3}}\right]$ · $\tau_{\mathrm{th}} \sim 1/\sqrt{\varepsilon}$

Gray LTE atmosphere (Lecture 9)

- $J = B = S$, $H = \sigma T_{\mathrm{eff}}^4/4\pi$
- Eddington: $T^4 = \frac{3}{4}T_{\mathrm{eff}}^4(\tau + \frac{2}{3})$, $T(0) = 0.841 T_{\mathrm{eff}}$, $T(2/3) = T_{\mathrm{eff}}$
- Exact: $T^4 = \frac{3}{4}T_{\mathrm{eff}}^4[\tau + q(\tau)]$, $q(0) = 1/\sqrt{3}$, $q(\infty) = 0.7104$, $T(0) = 0.811 T_{\mathrm{eff}}$
- Limb darkening: Eddington $I(\mu)/I(1) = (2+3\mu)/5$; exact $H(\mu)/H(1)$, $I(0)/I(1) = 0.344$
- Two-stream ($\mu = \pm 1/\sqrt{3}$): $T^4 = \frac{3}{4}T_{\mathrm{eff}}^4(\tau + 1/\sqrt{3})$
- Hydrostatic: $dP/d\tau = g/\kappa \Rightarrow P = g\tau/\kappa$

Lines (Lecture 10)

- $\int \sigma_\nu d\nu = \pi r_e c f \cdot \Delta\nu_D = \frac{\nu_0}{c}\sqrt{2kT/m + \xi^2}$ · Voigt $a = \gamma/4\pi\Delta\nu_D$
- Milne–Eddington (LTE): $\dfrac{F_\nu}{F_c} = \dfrac{1 + 2\beta/[3(1+\eta_\nu)]}{1 + 2\beta/3}$, maximum depth $1 - \frac{1}{1+2\beta/3}$
- Weak line: $W_\lambda = \pi r_e \lambda^2 N f$ · Curve of growth: $W \propto N \to \sqrt{\ln N} \to \sqrt{N}$
- NLTE: $S^{\mathrm{line}} = \frac{2h\nu^3}{c^2}\left[\frac{b_l}{b_u}e^{h\nu/kT} - 1\right]^{-1}$